Minimal Method Revision under Observational Equivalence: Failure-Governed Evolution without Self-Promotion

Working framework draft

August 2026

Abstract

A failed research or engineering episode rarely identifies the level at which a method should change: retrieval, environment, measurement, representation and method-basis defects can produce the same visible symptom. We formalize *minimal method revision under observational equivalence*. A latent situation determines an admissible minimal revision front, while a reviser sees only an evidence interface. Exact revision selection is possible if and only if that interface is constant on the required-decision fibres. Under arbitrary revision losses, the irreducible risk is the conditional Bayes risk inside those fibres. For a finite ambiguous fibre and finite registered intervention family, a panel resolves revision responsibility exactly if and only if it covers every observationally equivalent pair that requires a different revision; minimum-cost non-adaptive discrimination is therefore the weighted set-cover problem induced on those pairs. An adaptive programme is exact precisely when every terminal leaf is decision-pure. With stochastic adaptive transcripts, exact decoding exists if and only if the decision-class transcript laws are mutually singular; otherwise the best transcript rule retains the corresponding Bayes impurity. A protected-authority corollary shows that if internally identical transcripts can require opposite promotion decisions, no internal rule can be both sound and complete. These results motivate a failure-governed architecture with immutable adverse history, causal discrimination, isolated challengers, replay, protected fresh transfer, and external promotion custody. The formal results establish finite or declared probabilistic identification boundaries. Local hostile tests exercise registered implementation semantics only. A diagnostic glm-5.2 hidden-cause attribution archive scores 21/24 with three residual errors retained; this does not establish transferable self-improvement. P5-RD-01 and P5-RD-03 remain unexecuted. P5-RD-02 contributes only one public, local, known-fix development cluster and a post-outcome archival audit; it supplies no protected or fresh evidence and does not alter the original error. An outcome-blind SWE-agent execution preflight binds a native parser and isolation/resource envelope but leaves 12 C1 fields blocking. A separate MOSS preflight binds its parser, closed fallbacks, wallclock and two dependency locks but finds that the authoritative paper-linked source archive omits the native `benchmark/claw-eval` runtime; C2 therefore retains 14 blockers. A third preflight binds the DGM patch parser, empty fallback set and outer wallclock but correctly leaves its 23 unpinned dependency declarations unresolved, so C3 retains 15 blockers. A fourth preflight binds the ADIAS native parser, empty fallback set and foreground wallclock, but leaves dependency locking, bundled-task rights, rootless isolation and external custody unbound, so C4 also retains 15 blockers. A fifth preflight binds the Double Ratchet metric-only parser, isolated write surface, empty fallbacks, resource envelope and a 46-entry lock, but its native runner regenerates solver outputs and reports its development-locked metric each round; C5 retains 12 blockers. A sixth preflight binds ScienceClaw's exact source tree, a strict draft-shape parser and an outer wallclock, but its released CLI has no native revision selector and cannot close all model/tool fallbacks; C6 retains 16 blockers. A panel-wide successor then freezes one substantive, arm-neutral Defects4J Lang-1/Apache Commons Lang case core with exact Apache-2.0/CC0 rights and binds the identical candidate-visible bytes and task-content rights to all six arms. This

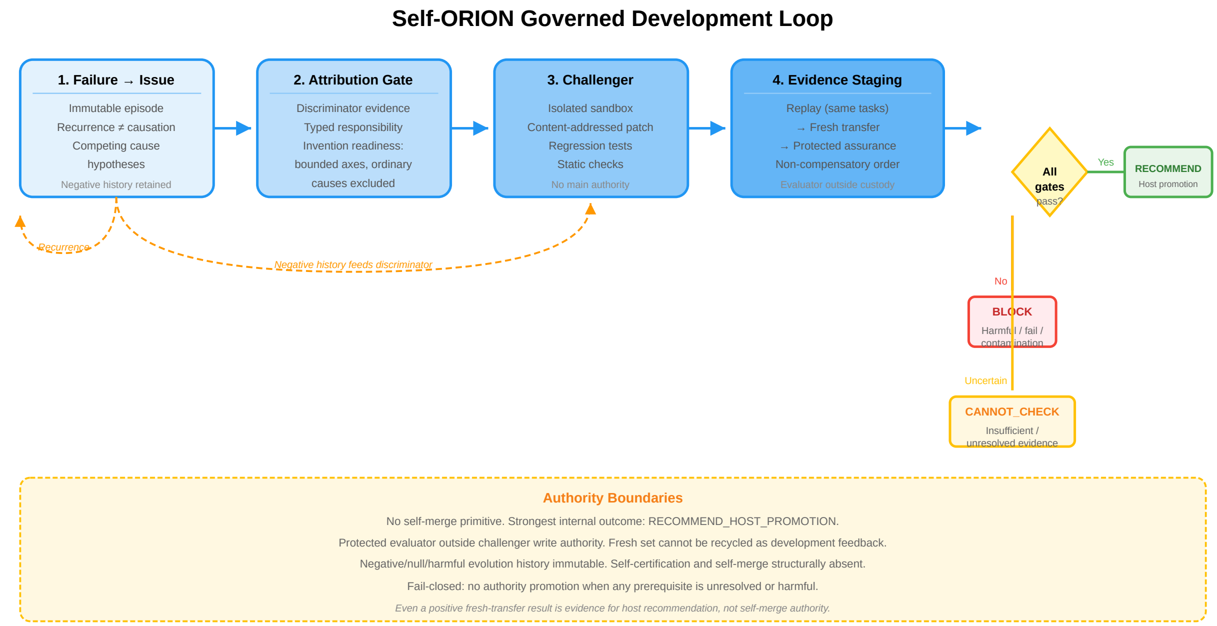

Figure 1: Self-ORION governed development loop. The failure-to-method fibre proceeds through persistent issue recording, causal discrimination, invention-readiness gating, isolated challenger execution, non-compensatory evidence staging (replay → fresh transfer → protected assurance), and a fail-closed authority boundary. The strongest internal outcome is a host recommendation; self-merge and self-certification are structurally absent.

closes 12 shared fields in one artifact, moving the panel from 42/126 to 54/126 bound and from 84 to 72 blockers. A byte-level native-environment successor then closes C1 with exact offline setup bytes and a fixed effective agent configuration, moving the panel to 55/126 bound and 71 blockers; five arm-native task environments remain unbound. A C3 byte-materialization successor excludes 1,595 prior-outcome files and constructs a 69-member lawful candidate seed, but the unchanged DGM cannot initialize without the excluded `initial/` performance metadata. A distinct static successor proves that zero-filling, removing self-edit action, or score-free parent selection would fabricate state or change the comparator. C3 therefore remains blocking and the aggregate census stays 55/126 bound and 71 blockers. Three deliberately separate C2 successors then close different obligations without changing that released-comparator census. V11 binds `runtime.task_environment` for an 8/21 successor; V12 binds exact candidate-visible bytes and task-content rights for a separate 9/21 successor; V13 binds complete scratch-image SBOM, licence and generated-artifact authority for a separate 8/21 successor. These states are non-aggregable and leave released MOSS at 7/21 bound and 14 blocking. Zero of six comparator arms is ready. A separately frozen 96-case revision-level panel was executed, but the subject remained at the safety floor and none of the seven registered terminals fired; that result grants no scientific authority. H1, H2 and governed improvement on protected fresh development tasks therefore remain undetermined, while the bounded H3/H4 checks are reported only as secondary panel observations rather than as general superiority evidence.

Keywords: minimal method revision; observational equivalence; active diagnosis; self-improving agents; fresh transfer; evaluator integrity.

1 Introduction

A method-evolving research system must answer two different questions: what change should be tried, and who is allowed to conclude that the change is better. Collapsing those questions creates a self-promotion channel. A coding agent may produce an impressive patch, but if it can also rewrite the benchmark, choose only favorable episodes, change protected governance, or certify its own evidence, improvement is not established.

Development is therefore treated here as an evidence-governed research process rather than as a direct self-edit loop. The claim under test is deliberately narrower than "autonomous self-improvement":

> Failures may become method knowledge only after persistent recording, causal discrimination, nearest-work/transfer challenge, isolated challenger execution, replay, fresh transfer and protected assurance; the candidate process cannot grant itself method authority or merge its own change.

A method state is not writable merely because a research run failed. A failure first changes the evidence available *about* the method. Only a separately governed transition can change which method is authorized.

1.1 Contributions

Four mechanisms carry the argument, and each is a property of the mechanism rather than of any particular system that implements it:

1. a failure-to-method path that separates observation, recurrence, attribution, lesson/challenger and authority;
2. non-self-promotional evolution, in which candidate generation and candidate authorization remain distinct;
3. protected fresh assurance, which requires evidence beyond the episodes used to construct the repair;
4. a separate authority layer whose final promotion remains in external host custody.

These are candidate differences from existing work, not established novelty or empirical superiority. Section 8 places each among its neighbours, and Section 11 states what the present evidence does and does not support.

2 Failure and null outcomes as persistent knowledge

Failed, blocked, partial and undetermined executions are stored as addressable experience rather than discarded as execution noise. A failure episode distinguishes the observed mode, propagated effect, task/variation identity, competing causes, detection evidence, recovery actions, residuals and falsifiers. Negative outcomes remain in history after later success.

This makes failure a potential research input, but not an automatic method lesson. Let e_j denote an immutable episode and $\phi(e_j)$ its observed failure signature. Repeated signatures can justify a recurrence hypothesis,

$$\phi(e_1) \approx \phi(e_2) \approx \cdots \approx \phi(e_n),$$

without justifying the causal statement that one mechanism caused every failure. Recurrence answers "does something structurally similar persist?"; attribution answers "which intervention should change the outcome?"

A valid development record therefore keeps failures, nulls and harmful interventions visible. Otherwise an adaptive system can manufacture an apparently improving history by retaining only successful challengers. Negative-history completeness is an assurance coordinate rather than an optional debugging convenience.

Where the record begins matters as much as what it holds. An execution record can expose a missing or failed coordinate of the method it was produced under; interpreting that coordinate as evidence *about* the method is a separate step, and it is deliberately non-automatic, because a missing specification, a provider outage, a data defect and a genuine operator defect call for different responses.

3 Minimal method revision under observational equivalence

The general object of this paper is not a particular self-editing agent. It is the decision problem created when several latent failures look the same to a method reviser but license different scientific changes. The relevant boundary is therefore informational before it is algorithmic: no more capable proposer, search procedure or acceptor can recover a distinction that its interface has erased.

3.1 Revision fronts and evidence interfaces

Let Θ be a set of latent development situations and let $(\mathcal{R}, \prec)$ be a well-founded strict partial order of candidate revisions. The interpretation of $r' \prec r$ is that r' changes strictly less of the method than r; representation repair and evaluator repair may be incomparable, so a total ladder is neither required nor generally appropriate. For $\theta \in \Theta$, let $\mathcal{A}(\theta) \subseteq \mathcal{R}$ be the revisions admissible after causal-responsibility, task-recovery, preservation, fresh-transfer, harm and authority constraints are applied, and let

$$\mathcal{M}(\theta) = \min_{\prec} \mathcal{A}(\theta)$$

be its minimal revision front. Well-foundedness guarantees a nonempty minimal front whenever $\mathcal{A}(\theta)$ is nonempty.

Let $Q(\theta)$ indicate a material residual. The required decision is the front-valued map

$$G(\theta) = \begin{cases} \mathsf{NO\_REVISION}, & \neg Q(\theta), \\ \mathsf{UNRESOLVED}, & Q(\theta) \text{ and } \mathcal{A}(\theta) = \varnothing, \\ \mathcal{M}(\theta), & Q(\theta) \text{ and } \mathcal{A}(\theta) \neq \varnothing. \end{cases}$$

Returning an antichain rather than silently choosing one incomparable element is deliberate. Additional licensed measurement may shrink a front when the plurality is caused by unresolved latent information. If several revisions remain minimal even with the situation fixed, measurement alone need not choose among them; an explicit downstream loss, host preference or retained antichain is then required.

A reviser observes an interface $\phi : \Theta \to \mathcal{Z}$, not θ itself. The interface can include a failure transcript, retrieved records, diagnostic labels, intervention results, issue history, and outputs of any number of donor systems.

Theorem 1 (revision-factorization theorem). There exists an exact revision rule $g : \phi(\Theta) \to \mathrm{im}(G)$ with $G = g \circ \phi$ if and only if

$$\phi(\theta) = \phi(\theta') \implies G(\theta) = G(\theta') \qquad \text{for all } \theta, \theta' \in \Theta.$$

Proof. If $G = g \circ \phi$, equal interface values necessarily give equal decisions. Conversely, if G is constant on each fibre of ϕ, define $g(z)$ to be $G(\theta)$ for any θ satisfying $\phi(\theta) = z$. Fibre constancy makes the definition independent of the representative. □

Theorem 1 is set-theoretic. Fibre constancy alone does not guarantee that the induced factor is measurable or computable. For a measurable reviser one must equip the spaces with sigma algebras and additionally require the induced g to be measurable on $\phi(\Theta)$. In the standard-Borel, finite-decision formulation below, measurable factorization is equivalently the requirement that G be $\sigma(\phi)$-measurable. The distributional result imposes these conditions rather than silently promoting the set-theoretic theorem.

Thus exact selection requires decision sufficiency, not recovery of every latent cause. A compressed interface is enough if it separates all situations that require different revision fronts. An arbitrarily rich interface is insufficient if even one such pair remains observationally equivalent. This is a decision-specific application of statistical experiment comparison [2]; neither the factorization argument nor Bayes decision theory below is claimed as a new general statistical principle.

3.2 Irreducible revision risk

For the distributional result, assume Θ and $\mathcal{Z}$ are standard Borel spaces and ϕ and G are measurable. Let $H \sim \mu$ on Θ, $Z = \phi(H)$, and $Y = G(H)$, with finite nonempty decision alphabet $\mathcal{Y}$ carrying its discrete sigma algebra. Let $\mathcal{D}$ be a finite nonempty action set that may include revision fronts and abstention, and let $L : \mathcal{D} \times \mathcal{Y} \to [0, \infty)$ encode the cost of under-revision, false broad revision, harmful transfer, or abstention. Write $p_y(z) = \Pr\{Y = y \mid Z = z\}$ for a regular conditional version, which exists under these assumptions.

Theorem 2 (mixed-fibre Bayes bound). Every measurable deterministic rule or measurable randomized Markov-kernel rule using only Z has risk at least

$$R^*(\phi, L) = \mathbb{E}_Z \left[\min_{a \in \mathcal{D}} \sum_{y \in \mathcal{Y}} L(a, y) p_y(Z) \right],$$

and a fibrewise Bayes action attains equality. Because both alphabets are finite, a fixed ordering supplies a measurable tie-break among minimizers. If $\mathcal{D} = \mathcal{Y}$ and $L(a, y) = \mathbf{1}\{a \neq y\}$, this becomes

$$R^*(\phi) = 1 - \mathbb{E}_Z \left[\max_y p_y(Z) \right].$$

Proof. Condition on $Z = z$. A randomized rule with probabilities $q(a \mid z)$ has conditional risk $\sum_a q(a \mid z) \sum_y L(a, y) p_y(z)$, a convex combination bounded below by the smallest constituent. Concentrating on a minimizing action attains the bound. Averaging over Z proves the first claim; zero–one loss yields the second expression. □

The bound makes asymmetry explicit. A policy that always chooses the broadest revision may have low under-revision loss but high false-broad-revision and harm loss; abstaining may be optimal on an impure fibre without making the fibre identifiable. No aggregate score can turn missing information into revision authority.

Corollary 1 (no sound-and-complete self-promotion under authority equivalence). Suppose two situations have the same complete internally visible transcript, but protected evidence authorizes promotion in one and forbids it in the other. No promotion rule using only that internal transcript can be both sound (never promote the forbidden case) and complete (promote the authorized case).

Proof. The two situations lie in one interface fibre but require different binary decisions, contradicting Theorem 1. □

External custody is therefore not a claim that an external evaluator is infallible. It is an information-and-authority requirement: the protected distinction must enter some decision interface not writable by the candidate. An always-abstain rule can remain sound, but it is not complete.

3.3 Discriminator design as a covering problem

Theorems 1–2 say when an interface is insufficient; the next result identifies exactly what a finite intervention programme must separate. Fix an observed value z and a finite ambiguous fibre $F_z = \{\theta : \phi(\theta) = z\}$. Let $\mathcal{T}$ be a finite set of candidate tests or controlled interventions, with deterministic protected outcomes $o_t(\theta)$ for $t \in \mathcal{T}$. Define the cross-decision pair universe

$$U_z = \big\{\{\theta, \theta'\} \subseteq F_z : G(\theta) \neq G(\theta')\big\}$$

and the subset distinguished by test t,

$$D_t = \big\{\{\theta, \theta'\} \in U_z : o_t(\theta) \neq o_t(\theta')\big\}.$$

Theorem 3 (discriminator-cover theorem). A non-adaptive test panel $S \subseteq \mathcal{T}$ permits exact revision selection on F_z if and only if

$$\bigcup_{t \in S} D_t = U_z.$$

Consequently, for test costs $c_t \geq 0$, whenever an exact panel exists, a minimum-cost exact panel exists and is exactly a weighted set-cover optimum of the finite induced instance $(U_z, \{D_t : t \in \mathcal{T}\}, c)$. If some pair in U_z belongs to no D_t, no panel drawn from $\mathcal{T}$ can resolve the revision decision.

Proof. The joint signature of θ under S is $(o_t(\theta))_{t \in S}$. It supports exact revision selection precisely when no two situations with different G values share a joint signature. A cross-decision pair has different joint signatures precisely when at least one selected test distinguishes it, which is precisely the stated cover condition. Minimizing the additive test cost over all such panels is the induced weighted set-cover problem. Finiteness of $\mathcal{T}$ makes its feasible family finite, so a nonnegative-cost optimum is attained whenever it is feasible. The last claim follows when a required pair is uncovered by the full candidate family. □

For an arbitrary candidate-test family, the cover equivalence remains valid for finite panels and the infimum exact-panel cost equals the corresponding weighted set-cover infimum, but an optimizer need not exist. For example, tests with one common required incidence pattern and costs $1/n$ have infimum zero without an attaining test. Any claim of a minimum beyond the finite setting therefore needs an explicit compactness or least-cost-representative condition.

Proposition 1 (adaptive leaf-purity criterion). For deterministic test outcomes, a terminating adaptive test policy permits an exact decision on F_z if and only if every reachable terminal leaf contains only situations with one common G value.

Proof. All situations reaching a terminal leaf generate the same observed test path. An exact terminal decision exists there if and only if their required decisions agree. Applying this condition to every reachable leaf proves both directions. □

These are structural equivalences in the classical minimum-test-set neighbourhood [15], not claims that every real experiment is deterministic or that a generic set-cover heuristic is statistically optimal.

Theorem 4 (stochastic adaptive-transcript criterion). Fix a possibly randomized terminating adaptive policy π and let T_π be its complete action–outcome transcript in a common standard-Borel transcript space. For each $\theta \in F_z$, let P_θ^π be the law of T_π. A measurable transcript decoder is exact almost surely for every situation in F_z if and only if there is a measurable partition $\{B_y : y \in G(F_z)\}$ such that

$$P_\theta^\pi(B_{G(\theta)}) = 1 \qquad \text{for every } \theta \in F_z.$$

For $F_{z,y} = \{\theta \in F_z : G(\theta) = y\}$, define the uniform class mixture

$$Q_y^\pi = |F_{z,y}|^{-1} \sum_{\theta \in F_{z,y}} P_\theta^\pi.$$

Equivalently, the finite family $\{Q_y^\pi : y \in G(F_z)\}$ is pairwise mutually singular. Under any prior $H \sim \nu$ on F_z, the smallest zero–one transcript-decoding error is

$$1 - \mathbb{E}_{T_\pi}\left[\max_{y \in G(F_z)} \Pr\{G(H) = y \mid T_\pi\}\right],$$

and a measurable fixed-order Bayes decoder attains it.

Proof. An exact decoder's inverse images give the displayed decision regions. Conversely, the partition defines an exact decoder. Since F_z and its decision image are finite, existence of that partition is equivalent to mutual singularity of the finite decision-class mixtures. The risk identity and attainment follow from Theorem 2 with T_π as the interface. □

In particular, adaptivity cannot exactly separate a cross-decision pair whose complete transcript laws under the fixed policy are identical. A stronger policy-uniform sufficient condition is equality of the history-conditional outcome kernels for every available intervention at every common reachable history; with world-independent policy randomization that may depend on observed history, this makes the complete transcript laws identical by induction for every policy. The converse need not hold for a fixed policy, because it may never select an intervention on which the kernels differ. Equality of one-test marginals is also insufficient: joint dependence can carry information. Optimizing risk plus experimental cost over an unrestricted adaptive-policy class may yield only an infimum; no optimal policy is asserted without a finite policy class or suitable compactness and lower-semicontinuity conditions. This connects method revision to active diagnosis and sequential experimental design [17, 7, 6, 9] while making the target the minimal authorized revision, rather than latent-cause recovery alone.

3.4 Controlled-state active revision with legal laboratory transitions

The stochastic transcript criterion fixes one policy. Choosing a policy also requires representing which discriminator remains legal after each outcome. For a finite nonempty ambiguous situation set W, let $G(w)$ be the required revision front and let $\mathcal{S}$ be a finite nonempty observable laboratory-state set. In state $s \in \mathcal{S}$, let $\mathcal{D}(s)$ be a finite nonempty terminal-action set (including revision, deferral, or unresolved actions) and let $\mathcal{E}(s)$ be a finite, possibly empty, legal intervention set. For $e \in \mathcal{E}(s)$, let $c(s, e) \geq 0$ be its cost, let $B(s, e)$ be a finite nonempty joint outcome–next-state branch set, and let $K_{s,e}(o, s' \mid w)$ be a known probability mass function on $B(s, e)$. The recorded state is assumed Markov-sufficient for intervention legality, costs, and these kernels. A destructive intervention moves to a state where it is no longer in $\mathcal{E}(s')$; a no-repeat restriction is therefore a transition law, not a prose instruction.

For terminal loss $L_s(d, G(w))$, set $\ell_{s,d}(w) = L_s(d, G(w))$ and define

$$\mathfrak{V}_{0,s} = \{\ell_{s,d} : d \in \mathcal{D}(s)\},$$

$$\mathfrak{V}_{n,s} = \mathfrak{V}_{0,s} \cup \bigcup_{e \in \mathcal{E}(s)} \Big\{ v : \quad v(w) = c(s,e) + \sum_{(o,s') \in B(s,e)} K_{s,e}(o, s' \mid w) v_{o,s'}(w), \\ v_{o,s'} \in \mathfrak{V}_{n-1,s'} \quad \text{for every } (o, s') \in B(s,e) \Big\}.$$

Stopping is permitted before the at-most-n budget is exhausted.

Theorem 5 (finite state-indexed revision frontier; donor-derived). The set $\mathfrak{V}_{n,s}$ is exactly the world-risk-vector set of all legal deterministic revision-discriminator trees from s. For an arbitrary nonempty fixed ex-ante credal set $\Pi \subseteq \Delta(W)$, deterministic and world-independent randomized perfect-recall minimax values are

$$\min_{v \in \mathfrak{V}_{n,s}} \sup_{p \in \Pi} p \cdot v, \qquad \min_{v \in \operatorname{conv}(\mathfrak{V}_{n,s})} \sup_{p \in \Pi} p \cdot v.$$

Both policy minima are attained, although an inner maximizing licensed prior need not exist when Π is nonclosed.

Proof. Backward induction enumerates stops and, for every legal root intervention, one depth-$(n-1)$ continuation per observed (o, s') branch. Conditional expectation gives the recurrence in one direction; attaching the inductively supplied continuation trees gives the converse. Finite perfect recall makes world-independent behavioural randomization outcome-law equivalent to a mixture over deterministic trees, hence the convex hull. The risk frontier is finite, its convex hull compact, and the credal support function is Lipschitz. □

With one prior, scalarization gives the ordinary controlled posterior Bellman equation, conditioning on outcome and observed next state. With a fixed nonrectangular prior set, the risk vectors must remain coupled until the root. A posterior-local adversary that independently pastes conditionals at different histories solves the rectangular path-law hull. For each fixed policy, its fixed-prior and rectangular robust expectations agree for every bounded path loss exactly when that policy's induced closed convex path-law family is already stable under positive-history pasting [8]. Equality of the optimized control values is weaker: it requires at least one fixed-prior-optimal policy whose value is not raised by rectangularization.

The recurrence and pasting boundary are inherited from finite POMDP, multiple-priors, and robust-control theory [21, 11, 16]; P5 does not claim a new alpha-vector or robust-POMDP theorem. Its use is claim-relative and fail-closed. An observed next state can be informative; an unavailable test cannot appear in a plan; randomization is only an executable scientific action if the protocol licenses it; and a zero-cost separating path defeats a positive acquisition- cost obligation. The nine retained failures S1–S9 remain separate successor research identities. The exact seven-witness receipt is a local arithmetic and exhaustive-tree check, not empirical evidence that any real discriminator kernel or laboratory state is valid.

3.5 The three retained errors as new experiments

The diagnostic archive contains three observed collisions between neighbouring revision levels. They are not evidence for Theorems 1–5 and are not repaired by renaming their labels. Each instead defines a successor pair universe and a factorial discriminator.

1. **Retrieval versus representation.** Hold the representation fixed while varying source access and retrieval policy, then hold retrieved information fixed while varying the representation. A retrieval-only rescue, a representation-only rescue, an interaction, and a joint null remain four distinct outcomes.

2. **Environment versus implementation.** Cross a content-addressed candidate with independently reconstructed environments. A code defect must follow candidate identity across environments; an environmental defect must follow environment identity across unchanged code. An interaction remains unresolved rather than being forced into either class.

3. **Representation versus method basis.** Cross a richer representation under a frozen method class with a method-class expansion under a frozen representation. Improvement only under the joint treatment identifies an interaction, not a license to credit either component alone.

P5-RD-01 and P5-RD-03 remain unexecuted. P5-RD-02 has only its immutable public-development V1/V2 observations and post-outcome V3 archival audit; those records neither correct P5-HC-012 nor supply protected freshness or H1–H4 authority.

These interventions refine the visible-symptom fibres that produced P5-HC-002, P5-HC-012 and P5-HC-018. The separate wide successor then asks whether the resulting decision interface transfers across source-disjoint project/issue families. A positive local discriminator is therefore a prerequisite for, not a substitute for, protected fresh-transfer evidence.

4 Causal discrimination and persistent issue state

A development issue persists across multiple investigations and interventions. A versioned development-issue record keeps a stable issue identity together with candidate causes, supported causes, discriminator evidence, failure episodes, intervention outcomes and lifecycle transitions. This prevents each repair attempt from becoming a new context that forgets why earlier alternatives failed.

For issue u, let

$$H(u) = \{h_1, h_2, \ldots, h_k\}$$

be competing cause hypotheses. A repair is not licensed merely because one hypothesis is narratively plausible. It is licensed only by discriminator evidence capable of separating the relevant alternatives. Where the discriminator cannot distinguish the hypotheses, the correct state is unresolved rather than an invented causal label.

This ordering is important:

$$\text{observed failure} \rightarrow \text{competing causes} \rightarrow \text{discriminator} \rightarrow \text{attribution} \rightarrow \text{candidate intervention.}$$

Moving the intervention before the discriminator encourages post-hoc explanations of whatever the candidate happened to change.

Persistent issue state also makes negative interventions scientifically useful. A harmful or null repair can narrow the supported cause set, expose a missing interaction, or invalidate an earlier lesson. Such episodes remain attached to the issue rather than being deleted once a later challenger succeeds.

5 Nearest-work challenge, challenger archives, and invention readiness

Repeated failure is not read as an instruction to invent a new operator. Before structural invention, ordinary causes, known mechanisms, transfer candidates and mature parent-domain treatments are searched for first. The invention-readiness gate asks whether relevant search axes are bounded-flat, ordinary causes have been excluded to the supported extent, the residual survives distinct variations, and discriminator evidence identifies a missing-structure class rather than a generic score deficit.

Candidate method changes are retained as challengers rather than silently overwriting the incumbent. The archive records candidate identity, generating issue/evidence, attempted intervention, execution result, regression result, fresh-transfer result, protected-assurance outcome and final host disposition. Failed and harmful candidates remain queryable.

This structure absorbs useful mechanisms from evolutionary and self-improving-agent work without inheriting self-authorization. Archives, iterative candidate generation, population search and evaluator-guided optimization may all be valuable proposal mechanisms. They become method authority only through the separate assurance path.

5.1 Method-level challengers and method-space expansion

A method challenger is a candidate structural change linked to documented failure or obstruction records, not a claim that a new method has been invented. The versioned method-challenger record binds the subject revision, generating failures, competing ordinary causes already challenged, known-method search routes and their inadequacy, assimilated donor mechanisms, the structural edit and its predeclared discriminator. Candidate generation is therefore separate from host adoption. A candidate must traverse

$$\begin{aligned}\textsc{Diagnose} &\rightarrow \textsc{Search/Absorb} \rightarrow \textsc{Challenger} \rightarrow \textsc{IsolatedRun} \\ &\rightarrow \textsc{Replay} \rightarrow \textsc{Fresh} \rightarrow \textsc{Protected} \rightarrow \textsc{HostDisposition}.\end{aligned}$$

A replay improvement cannot compensate for harmful fresh transfer, and missing protected evidence blocks disposition. Rejected, null and harmful histories remain append-only and queryable; neither a generator nor a challenger can rewrite the evaluator or authority registry. For example, a structural adaptation that improves its motivating replay but harms a protected family on fresh transfer remains rejected, even if its development score is higher. Known-method search and transfer must be challenged before an invention gap is declared; repeated failure alone never makes a method ready for invention. The host alone may adopt or recommend adoption.

This is a governance boundary, not an invention result. The mechanism can govern whether a method candidate is safely evaluated and presented to a host; whether a learned system can generate valuable new method structures is a different question, and nothing here bears on it.

5.2 Transferred design knowledge

A prior specification audit re-examined the mature method surfaces this design inherits, treating them as prior design knowledge rather than as results. Fourteen provisional structural dimensions across 59 registered mechanics produced 826 explicit transfer questions; evidence-bound answer records moved that registered structural frontier from 826 to 0 with attributable answer and audit closure.

This result is deliberately narrow:

$$\text{specification closure} \neq \text{empirical validity} \neq \text{promotion authority.}$$

It can reduce rediscovery of already specified mechanics and expose remaining empirical work, but it cannot establish that the transferred contracts improve behaviour on fresh tasks.

6 Isolated challenger execution and protected change control

The development controller composes a governed path of the form

$$\begin{aligned}&\text{persistent issue/evidence} \to \text{research and nearest-work challenge} \to \text{candidate proposal}\\&\to \text{content-addressed patch} \to \text{isolated sandbox execution} \to \text{regression/fresh assurance}\\&\to \text{append-only evolution record} \to \text{host recommendation.}\end{aligned}$$

A language model, coding agent, program synthesizer or other worker may generate the proposed change. It is not the controller and does not own the evaluator. The proposed bytes are content-addressed before execution so the evaluated candidate is the candidate later presented for promotion.

Protected evaluator, registry and governance paths are hard rejection coordinates. A candidate that improves visible scores while modifying its own protected evaluation surface is not treated as an improved method. Candidate execution occurs behind a host-owned isolation boundary, and the change controller exposes no merge primitive.

The strongest internally representable successful outcome is therefore a recommendation that the host promote the candidate, not "self-merged" or "governed by self declaration." This is a structural authority boundary: even a perfect candidate score cannot compensate for missing or compromised change-control evidence.

7 Replay, fresh transfer, and protected assurance

A challenger can overfit the exact failure episodes used to construct it. Three evidence roles are therefore kept separate.

Replay. The candidate must address the supporting failure under a bound replay rather than merely changing a proxy score.

Fresh transfer. The candidate must be evaluated on distinct task, split, run/evaluation and variation identities not used as the repair's support set. Improvement on visible development evidence combined with regression on fresh assurance is represented explicitly as meta-overfitting rather than averaged into a favorable mean.

Protected assurance. The authority-bearing evaluator is outside candidate write authority and binds the exact candidate, subject revision, evidence artifact, evaluator identity, lineage, epoch and split. Self-verification, post-hoc evaluator evidence, non-fresh evidence, mixed subject/evaluator epochs and unresolved checks block promotion.

The readiness correction is intentionally asymmetric. Even a complete independent set of passing checks can reach only a readiness state that is still pending external attestation; final promotion stays outside the evaluated system's own custody. This prevents a boolean checklist, a candidate-generated receipt, or a self-selected benchmark from manufacturing governed method authority.

Negative history is part of assurance. A promotion packet that omits failed or harmful alternatives can give a false picture of selection pressure and regression risk; therefore the archive and external handoff must preserve those outcomes.

8 Related work

The neighbouring literature is strongest exactly where this paper does not compete. Every mechanism discussed below is treated as a component the design may absorb, not as a rival: a stronger proposer, attributor or acceptor makes the authority boundary drawn here more useful, not less.

Archives and self-modification came first. ADAS automates agent-system design through a meta-agent and an archive [10]; Darwin Gödel Machine performs open-ended self-code modification with empirical validation, an archive and sandboxing [25]; and ADIAS makes persistent issue identity, lifecycle, evidence and intervention-outcome history a first-class optimization state [12]. Persistent issue state, an archive of candidates and sandboxed validation are therefore inherited here rather than introduced.

Failure attribution is a second neighbourhood, and a crowded one. SAGE uses multi-hypothesis failure attribution to generate evidence-grounded explanations and routes verified root causes to intervention levels [14]. CausalFlow uses counterfactual intervention to estimate causal responsibility and to generate minimal repairs [3]. Failure-driven self-improvement converts failed computer-use trajectories into diagnosed code-level improvements [22]. Each of these answers *which cause explains the failure*. The question they leave open is what an attributed cause is then permitted to authorize, and on whose evidence that authorization rests.

Acceptance-specific work removes another possible novelty route. PACE casts repeated incumbent-versus-candidate commits under adaptive self-evolution as anytime-valid paired sequential testing, so optional-stopping-safe commit acceptance is not a P5 novelty [20]. SEA independently confines modification around a frozen base and admits changes through anytime-valid certificates [18]. P5 therefore treats acceptance statistics and verifier gates as absorbed mechanism pressure, not as its residual contribution.

The broader recursive-self-improvement literature raises the bar once more. A 2026 survey of 1,250 recent papers separates bounded self-refinement from stronger recursive self-improvement and places self-evaluation on a verification hierarchy, with demonstrated improvement strength tracking stronger external or formal verification [5]. PAST-Bench independently asks whether retained experience actually causes later-task improvement under matched experience-on/off conditions and whether the gain follows the intended save–retrieve–update pathway [23]. SEVA reports that repeated self-evolution can produce benchmark specialists that improve on one benchmark while regressing on another [24]. These results make generic claims such as "the system learns from failure", "the system improves itself", "the candidate passed a verifier", or "experience caused later gains" mature targets rather than a defensible P5 breakthrough.

The surviving P5 residual is a different scientific question: *who has authority to turn an internally generated diagnosis or proposal into an adopted scientific method?* Self-ORION separates proposal authority from adoption authority. Its internal controller may diagnose, nominate a revision, select a computation, recommend containment or compose a next-step report, but the registered control object grants none of those outputs revision, adoption, promotion, merge or global-stop authority. The three retained hidden-cause errors therefore become a non-vacuous test condition for the architecture: fallible internal diagnosis can guide investigation without becoming a self-issued adoption certificate. This bounded non-self-promotion result is the paper's present contribution. Transferable fresh-task improvement against strong self-evolving baselines remains a separate prospective extension and is not required to establish the authority-separation theorem. Self-evolving coding systems press on the same boundary from the engineering side. MOSS rewrites agent source from production-failure evidence, validates candidate images by replay, and places deployment behind user consent and rollback checks [4]. Ratchet manages a self-authored skill library with outcome-driven retirement and bounded capacity while measuring held-out transfer [26], and Verifier-as-Gatekeeper shows that harmful skill admission contaminates later evolution, testing heterogeneous pre-commit gating together with frozen-pool transfer [19]. MEGA accumulates durable optimization assets in a typed wisdom graph and uses controlled evaluation to attribute improvement effects across optimization cycles [13], while RewardHackingAgents makes evaluator tampering and held-out leakage measurable through patch and access telemetry and evaluator locking [1]. A current survey of self-evolving coding agents treats feedback reliability, benchmark overfitting, safety, maintainability, cost and generalization as field-level concerns [27]; the position taken here is that those concerns are to be measured, not cited as motivation.

Acceptance statistics form a third neighbourhood. PACE casts repeated incumbent-versus-candidate commits under adaptive self-evolution as anytime-valid paired sequential testing [20], and SEA admits self-modifications through anytime-valid certificates around a frozen base [18]. An anytime-valid acceptor answers when an observed difference is real under optional stopping. It does not answer which evidence the difference is allowed to be computed over, and it is adopted here as a within-stage device rather than as the acceptance rule itself.

Finally, transfer measurement supplies the outcome discipline. PAST-Bench evaluates whether retained experience causes later-task improvement under matched experience-on/off conditions [23], which motivates pathway-sensitive evaluation rather than treating memory presence as learning. SEVA reports that repeated self-evolution can produce benchmark specialists that improve on one benchmark while regressing on another [24], which is why replay evidence and fresh-transfer evidence are measured and reported separately below rather than averaged.

What remains after all of this is narrower and explicitly compositional. A persistent development issue may license a method change only after evidence-bound causal discrimination and invention-readiness checks; acceptance additionally requires non-compensatory staged evidence in the order static checks, replay, independent protected fresh transfer and non-harm, and protected assurance, with immutable retention of negative, null and harmful history and of compromise telemetry, with evaluator and holdout custody outside challenger write authority, and with self-certification and merge authority structurally absent. Promotion remains a host decision. Whether that composition yields more transferable and less compromise-prone improvement than simpler self-edit or acceptance systems is an empirical question; architecture alone cannot settle it, and the evidence reported here does not.

9 Methods

The design choices in this section were fixed before any result-bearing campaign. Machine-readable artifacts govern rather than this narrative: a frozen design, a hidden-cause case schema, a replay/freshness and harm-accounting policy, and a staged acceptance policy. Each is archived, and Section 12 names where, together with the preregistration identifier under which the design was frozen.

9.1 Three-valued outcomes

Every decision this paper measures is three-valued rather than binary. A governance check may be determined to hold, determined to fail, or left *undetermined*; a candidate repair may be determined to transfer, determined to harm, or left undetermined. An outcome that remains undetermined is recorded and reported separately from an outcome determined to be false, and the two are never merged. The distinction is the point: an unbound execution identity, an absent credential, a protected evaluator that does not yet exist, or a campaign that has not been run all produce undetermined outcomes, and an undetermined outcome is not evidence of absence. Throughout this paper an outcome described as *undetermined* is this third value; it is carried in the evidence record and in every downstream count, and missing protected evidence is never silently resolved to zero or to a negative. The published tables print the same third value under its recorded status name, so that a table still awaiting its campaign cannot be misread as a table reporting that nothing happened.

9.2 The implementation under test

The implementation evaluated here is the Self-ORION development controller; it is the artifact under test. The rest of the paper describes the governance mechanism rather than the product, so that each claim reads as a claim about the rule and not about a named system. Source, protocols and evidence archives are listed in Section 12.

9.3 Local hostile falsifier suite

The current local suite exercises the governance semantics rather than claiming open-world improvement. Registered cases include recurrence without causality, competing-cause discrimination, negative intervention retention, replay/fresh-transfer separation, evidence-bound reopening, invention-readiness constraints and protected-path reward-hacking attacks. A candidate that modifies protected registry or governance state is rejected even when visible and fresh score deltas are constructed to look perfect.

These tests support the statement that the implementation can represent and refuse the registered failure modes. They do not show that it discovers correct causes or produces beneficial repairs on real development problems.

9.4 Prospective external comparison

The outcome-blind design is frozen in advance. The primary hypothesis is that the full governed path produces greater *protected fresh-task improvement* than the strongest matched self-edit or self-evolution baseline on hidden-cause development cases, while the harmful-transfer rate remains within a predeclared two-percentage-point non-inferiority guard. The design target for fresh improvement is an absolute five-percentage-point advantage. These values are prospective margins, not observed effects.

Registered hidden-cause families are retrieval miss, routing or planning miss, implementation bug, environment/dependency/tool failure, evaluator or metric bug, representation gap, measurement or specification gap, and genuine method-basis gap. The candidate sees the symptom and context but not the protected root-cause label. Each episode separates motivating evidence, replay tasks and an independently frozen fresh-transfer set that changes at least one task, domain, model, environment, data or tool axis.

Strong baselines include a fixed agent, direct self-edit, meta-agent design search, archive-based self-edit, issue-centric optimization, multi-hypothesis attribution, counterfactual repair, failure-driven patching and evaluator-only evolutionary search where runnable. Ablations remove persistent issue state, causal discrimination, invention readiness, independent fresh transfer, protected evaluator custody, negative history and access telemetry; a self-certification or self-promotion ablation is permitted only in an isolated benchmark safety lane and cannot affect the production authority path.

Primary outcomes are protected fresh-task improvement, fresh-transfer success and harmful-transfer rate. Secondary outcomes include cause macro-F1, false method-change rate, intervention success conditional on attribution, evaluator, holdout and negative-history compromise, correct blocking of an undecidable case, recognized-failure recurrence, negative-history completeness, motivating-task improvement and resource cost. Wilson intervals are used for standalone rates and paired percentile bootstrap intervals for matched improvement differences; stochastic repetitions are nested within episode and system. Harmful tail and family regressions are reported separately rather than averaged away.

9.5 Design freeze and fresh-transfer custody

The design is frozen and the freeze records that no outcome had been accessed when it was cut. This is not yet an execution freeze. The exact subject revision, hidden-cause suite hash, motivating/replay/fresh split hashes, provider and model versions, baseline configurations, evaluator artifact and evaluation epoch all remain unbound. The fresh evaluator and holdout must be frozen outside challenger write authority before candidate generation, and the fresh set cannot be recycled as development feedback without losing its freshness. Search, file, evaluator and patch access is retained for contamination and integrity review. The external transfer benchmark used as a reference point is pinned to a specific upstream revision, recorded with the protocol; any actual use of it still requires a frozen task and split manifest.

9.6 Statistical methods

Standalone rates use Wilson 95% intervals. Matched improvement differences, when an archive exists, use paired percentile bootstrap with 10 000 resamples nested in episode $\times$ system $\times$ seed. The primary hypothesis family is protected fresh-task improvement with a harmful-transfer guard; Holm correction applies to inferential secondary attribution and governance families. Catastrophic fresh regressions are reported by family and are not averaged away. Missing protected evidence remains undetermined and is never entered as zero. The diagnostic 21/24 attribution interval in Table 1 is a standalone Wilson interval on $n = 24$; it is not a primary effect size.

9.7 Threat model

Evaluator compromise, holdout leakage, search contamination, replay-only laundering, missing-stage compensation, negative-history deletion and self-certification are treated as first-class failure modes rather than as missing data. A candidate that can rewrite the protected evaluator, recycle the fresh set as development feedback, or certify its own readiness has not produced an admissible improvement. Local hostile tests exercise registered instances of those attacks; unknown gaming strategies remain outside the present evidence.

9.8 Revision-level successor design and frozen execution

A successor design adds a narrower formal safeguard ahead of any broader self-revision experiment. Competing revision-responsibility hypotheses may remain ambiguous or unresolved, and required interface-adequacy checks may block escalation from an observed failure to a model, method or objective rewrite while a narrower observation, measurement or representation repair remains indicated. The implementation can nominate a claim-relative minimal candidate mechanic for later testing, but the responsibility report, the interface report and the revision gate are all structurally non-authorizing: they do not replace the invention-readiness gate and do not grant adoption or promotion.

A second tranche makes three further distinctions explicit without granting new operating authority. Internal computation is itself represented as a costed epistemic action, so a hard verification obligation can require an expensive check even when a cheaper action has greater scalar expected value, while optional non-positive computation may stop locally. An uncertainty-containment envelope can restrict the contexts in which a model or claim is admissible without rewriting the model or certifying it as correct. And social corroboration is evidence-bound: different worker or verifier identities are not counted as independent when their registered upstream sources or direct observations overlap, and unresolved hidden contributors prevent exclusive causal credit. These are implementation contracts for later countermodels and experiments, not evidence that adaptive computation, containment or social-state tracking improves any outcome.

The successor experiment is a separate, protected-custody revision-level benchmark rather than a rewrite of the frozen design above. Its cause-confusable panel exposes the same candidate-visible symptom under different protected revision responsibilities; it evaluates matched no-revision, broad-self-edit, method-open-only, world-model, representation-regime, generic-diagnosis and fully composed policies, and it includes feedback-permutation and no-feedback controls. The final subject, 96-case suite, split, evaluator and baseline bindings were verified before execution. Under the unchanged one-probe policy, however, each panel hypothesis registered two expected diagnostics; the responsibility gate therefore retained an unresolved discriminator after the single observation and the fully composed subject never promoted. This execution is reported as an adverse identification result, not retuned into a positive terminal.

9.9 Authority boundary

Whether governed method evolution improves fresh development outcomes remains undetermined because the frozen execution reached no registered terminal: FULL_T7 accuracy and fresh-transfer success were both 0.125, at the safety floor, while generic causal diagnosis reached 1.0 on the same constructed panel. Repository continuous integration, structural specification closure, local hostile tests, replay-only success, a diagnostic 21/24 attribution score, and bounded secondary governance checks are each incapable of resolving the broader boundary. Even a future positive fresh-transfer result would be evidence for an external host recommendation rather than for self-merge authority.

10 Results: the diagnostic attribution archive

The primary diagnostic archive in this revision is a single-model hidden-cause attribution run over 24 constructed cases, three in each of the eight registered families, produced with the glm-5.2 model. The metrics below are recomputed from the raw per-case records; the sidecar report is checked against those rows and is refused if it disagrees.

Accuracy is $21/24 = 0.875$, with the nominal Wilson score interval given in Table 1. The interval treats the 24 fixed constructed cases as Bernoulli units; because they are not a probability sample and come from one model run, it is not a population confidence interval. This is a descriptive finite-suite diagnostic score, not protected fresh-task improvement, a matched baseline comparison, or a test of the primary hypothesis.

Three residual errors are retained and are not relabeled as successes (Table 2). Case P5-HC-002, whose protected cause was a retrieval miss, was attributed to a representation gap at medium confidence; case P5-HC-012, an environment, dependency or tool failure, was attributed to an implementation bug at high confidence; and case P5-HC-018, a representation gap, was attributed to a method-basis gap at high confidence.

Each error now has a distinct successor research identity. P5-RD-01 crosses retrieval access with representation regime; P5-RD-02 crosses candidate bytes with independently reconstructed environments; and P5-RD-03 crosses representation expansion with method-class expansion. Their four cells retain main effects, interaction, and joint-null outcomes separately. P5-RD-01 and P5-RD-03 have not been executed. P5-RD-02 now has three immutable public-development identities; none changes the diagnostic error or contributes corrected attribution accuracy.

`P5-RD-02.DEFECTS4J.LANG1.V1` ended CANNOT_CHECK with `HARMFUL_OR_ADVERSE_CELL_PRESENT`. Its registered JDK 17 arm violated Defects4J's Java 11 execution contract, and its raw-upstream-tree identity model incorrectly rejected the prepared local commits and trees that Defects4J creates during checkout. V2 was therefore preregistered as a materially new successor, not used to rewrite V1. It replayed the public known fix for Defects4J Lang-1 as one public development cluster across two content-addressed Java 11 distributions. All four V2 cells were valid: the buggy revision failed the registered developer test in both environments and the known fixed revision passed in both. Thus the registered contrasts were `implementation_effect`=1.0, `environment_effect`=0.0 and `interaction`=0, giving the primary terminal `IMPLEMENTATION_MAIN_EFFECT` with the separately retained harm modifier `NO_ADVERSE_CELL`. These are exact registered contrasts within one source/bug/fix cluster, not an estimated population main effect; the two Java distributions and four cells are within-cluster conditions rather than independent scientific units.

A direct retention audit then found a provenance limitation in V2. Its runner used `Path.with_suffix` for both checkout and test phase stems, collapsing the two phases to the same stdout/stderr paths in each cell. The retained files validate all eight declared test-stream hashes, but none of the eight declared checkout-stream hashes; those checkout bytes were overwritten and are not recoverable from the V2 archive. This limitation does not rewrite the registered V2 cell outcomes or terminal.

Because V2 outcomes were already known, `P5-RD-02.DEFECTS4J.LANG1.V3` was frozen as an audit-only archival successor rather than an independent replication. Its four cells replayed the same local pattern, while all 16 phase-qualified stream files matched their declared hashes and every cell retained four distinct phase paths. Its `ARCHIVAL_REPLAY_COMPLETE` terminal means only that this successor preserved its own replay logs. It neither recovers the missing V2 checkout bytes nor adds confirmatory causal, generalization or protected authority beyond its own archival-integrity result.

The scientific unit is one public Defects4J Lang-1 source/bug/fix cluster under a local evaluator (cluster-level $n = 1$), not the four cells, two Java distributions, or sixteen V3 phase streams. It is a known-fix replay, not a generated repair or a protected campaign, and establishes no population generality, independent custody, external independence, promotion authority, protected transfer or protected freshness. H1–H4 remain CANNOT_CHECK, and the bridge remains at P5_BRIDGE_PUBLIC_DEVELOPMENT_BOUND__PROTECTED_PROVIDER_AND_SIX_ARM_EXECUTION_CANNOT_CHECK. The three successor studies still serve to test whether registered interventions split the mixed revision fibres identified in Section 3; if a protected pair survives every registered intervention, the correct output remains UNRESOLVED.

Terminal-adapter interface result

The V2 source-interface audit supplies a formal result, not a comparator outcome. Six prospectively frozen same-visible-symptom pairs show that each raw native status fibre contains at least two minimal revision classes; accordingly zero raw native symptoms license a singleton class. Its deterministic synthetic suite passes 90/90 contract cases, with 70 UNRESOLVED terminals and 20 certificate/write-surface proposed-action types. The V3 successor does not relax that gate: it passes 231/231 declared fictional cases, with 40 records in 20 supported constant fibres and 191 UNRESOLVED; raw-native licences remain zero and ScienceClaw supports no singleton. No comparator, native output example, public or protected outcome, or performance table was used. The conditional support counts (seven, four, four, four, one, and zero across the six fixed arms) describe interface coverage only; the 231 rows are neither independent units nor correct-repair observations. The execution ledger retains 108 blocker instances, 18 per arm. Zero of six remains confirmatory-ready, and H1–H4 remain CANNOT_CHECK.

The later outcome-blind C1 V4 preflight binds six SWE-agent execution fields, making C1 9/21 bound and 12/21 blocking. It does not parse a scientific result: authored synthetic native-terminal checks all remain UNRESOLVED at the revision-class layer, and the isolation wrapper refuses execution with all 12 blockers. The other five arms remain unchanged; at this C1-only snapshot the merged panel has 102 blocker instances, zero ready arms and no performance or superiority observation.

The distinct outcome-blind C2 MOSS V4 preflight binds four fields beyond its three prior source identities, making C2 7/21 bound and 14/21 blocking. Its schema-v6 parser preserves native status/verdict but emits UNRESOLVED; exact host and 1,196-entry OpenClaw locks are retained. The complete authoritative tree contains zero `benchmark/` paths even though the native trial runner requires `benchmark/claw-eval`, so a full native smoke is CannotCheck. Updating C2 changes the current panel count from 102 to 98 blocker instances, not to a performance observation: zero of six remains ready and H1–H4 remain CANNOT_CHECK.

The distinct C3 DGM V4 preflight binds a closed native patch parser, no fallbacks and a 21,600-second outer watchdog, making C3 6/21 bound and 15/21 blocking. Its 23 requirements declarations contain zero exact pins and are not mislabelled as a resolved dependency lock. The synthetic PATCH_CAPTURED fixture retains native completion but maps to UNRESOLVED; a protected-key fixture is refused. Replacing C3's 18 V3 blockers with 15 changes the current panel count from 98 to 95. Zero of six remains ready and H1–H4 remain CANNOT_CHECK; no native DGM, model, task, benchmark or protected outcome was run.

The distinct C4 ADIAS V4 preflight binds a hashed report parser, an empty fallback/search policy and a 21,600-second foreground watchdog, making C4 6/21 bound and 15/21 blocking. The 1,578-file source manifest is byte-addressed, but four VCS dependencies are unpinned, the mutable host-networked container path does not isolate provider keys, and bundled task rights lack component licence/NOTICE files under `data/`. Replacing C4's 18 V3 blockers with 15 changes the current panel count from 95 to 92. Zero of six remains ready and H1–H4 remain CANNOT_CHECK;

no native ADIAS model, task, benchmark or protected outcome was run.

The distinct C5 Double Ratchet metric-only V4 preflight binds six fields beyond the three source identities, making C5 9/21 bound and 12/21 blocking. Its results-only write surface, resource envelope and 46-entry uv lock are outcome-blind interface bindings. The official runner nevertheless regenerates hosted solver outputs and reports its development-locked metric every round; the tree contains no frozen split/result payload and no P5-native task surface. Replacing C5's 18 V3 blockers with 12 changes the panel count from 92 to 86. Zero of six remains ready and H1–H4 remain CANNOT_CHECK; no native model, benchmark, P5 case, protected panel or performance outcome was run.

The distinct C6 ScienceClaw V4 preflight binds two fields beyond its three source identities, making C6 5/21 bound and 16/21 blocking. The strict native-shaped draft parser preserves every fibre as `UNRESOLVED`, and an outer process-group wallclock is bound. The released CLI provides no native revision selector, nominal dry-run would execute an investigation, and the model/tool/reasoning/skill/dependency fallbacks cannot all be disabled. Replacing C6's 18 V3 blockers with 16 changes the panel count from 86 to 84. Zero of six remains ready and H1–H4 remain CANNOT_CHECK; no ScienceClaw model, tool, service, benchmark, protected outcome or prior-result payload was executed or decoded.

The subsequent panel-wide V5 audit changes no arm result. Direct recomputation gives 42 bound and 84 blocking field instances out of 126, with zero of six arms ready. Fourteen exact blocker equivalence classes collapse into five evidence-producing work packages of sizes 25, 18, 18, 15 and eight. This is a 79-item reduction in the coordination queue (84 to five), not a reduction in the 84 scientific blocker instances: V5 creates zero new field bindings and no effect estimate. It also exposes an unmatched resource contract despite all six wallclock fields being individually bound. Its outcome-blind local validator passes 74 checks and re-verifies six parser digests, but those checks authorize neither execution nor performance. Fresh packet assurance is not uniform: C4's native validator rewrites a temporary-path-dependent audit hash, so its frozen audit receipt is determined not to reproduce. The cleaned C6 packet lacks the `.source-audit` checkout required by its native validator, so fresh C6 source validation remains CannotCheck. Both adverse assurance findings are retained rather than suppressed.

The P5 V6 common-case successor is the first material closure produced by that root registry. It freezes one complete rights-cleared Lang-1 candidate core and six acceptance receipts, creating exactly two new bindings per arm: common candidate-visible case bytes and common task/benchmark-content rights. The panel therefore moves from 42/126 to 54/126 bound and from 84 to 72 blocking. The result does not infer six arm environments from one source archive: all six `runtime.task_environment` fields remain blocking, zero arm is ready, and H1–H4 remain CANNOT_CHECK. The packet-local validator reports 741/741 checks and verifies 22 checksum entries without executing a model or opening an outcome.

The V7 native-environment fan-out closes one further field without executing an arm. Exact offline setup bytes and a fixed effective agent configuration bind C1's `runtime.task_environment`; the other five receipts remain blocking because their arm-specific artifacts do not exist. The aggregate census is therefore 55/126 bound and 71/126 blocking, with five R2 instances and zero of six arms ready. The packet validator reports one environment bound, five blocking, 67 byte references verified and 25 packet files hashed.

The C3 V8/V9 successors close no further field. V8 builds a lawful 69-member candidate seed and excludes 1,595 prior-outcome files, but the unchanged native initializer requires the excluded `initial/overall_performance` state. V9's one static exact-byte/AST gate proves that an outcome-free adapter would either fabricate performance, restore excluded outcomes, eliminate the nondegenerate self-edit action or replace native parent-selection semantics. The C3 environment remains blocking, so the aggregate stays 55/126 bound, 71/126 blocking and 0/6 ready. No DGM, model, benchmark, scorer or outcome is executed.

Three later C2 packets test distinct lawful-successor obligations and must not be read as a cumulative version chain. V11 binds `runtime.task_environment` for the separately authored `C2_LAWFUL_NATIVE_BYTE_SUCCESSOR__ORION_V11`, yielding 8/21 bound and 13 blocking on the C2 V4 count basis. Its six registered public hexadecimal classification classes pass the successor gate, but the task and evaluator are authored post-outcome development instruments rather than released MOSS or a protected panel. V12 separately binds `inputs.candidate_visible_case_bytes` and `rights.task_and_benchmark_content` for `C2_SOURCE_NATIVE_VISIBLE_CORE_SUCCESSOR__ORION_V12`, yielding 9/21 bound and 12 blocking. V12 does not inherit V11's runtime field.

V13 separately closes only `rights.container_and_generated_artifacts` for `C2_RIGHTS_CLEARED_SCRATCH_IMAGE_SUCCESSOR__ORION_V13`, yielding 8/21 bound and 13 blocking. Its exported Linux/arm64 scratch layer contains exactly nine regular files and no symlinks. The normalized frozen-rootfs, exported-layer, file-level-rights and SPDX 2.3 file sets are identical; full Apache-2.0/NOTICE/CC0 bytes and explicit generated-artifact retention, disclosure and publication authority accompany the image. The frozen runtime receipt records exact stdout `ORION V13 container pass\n`, exit 0, no network, read-only rootfs, all capabilities dropped, no-new-privileges and an empty container diff. The image was archived, removed from the daemon and the ephemeral remote build environment was destroyed.

None of V11–V13 inherits either other state, and their field closures cannot be unioned. Released MOSS remains 7/21 bound and 14 blocking, and the separate successors do not alter the authoritative 55/126 panel census. Consequently 0/6 arms remain ready and H1–H4, performance and superiority remain CANNOT_CHECK. No MOSS model, protected case, evaluator, scorer or performance outcome was executed by these packets.

Frozen revision-level panel: no registered terminal

The additive revision-level protocol verified its bound subject, protected suite, split, evaluator and baseline identities and then executed ten frozen policies on 96 constructed confirmatory cases, with 48 cases in each recorded half. FULL_T7 achieved revision-label accuracy 0.125 — **12/96** cases — false-broad rate 0, harmful-regression rate 0 and fresh-transfer success 0.125. Generic causal diagnosis achieved accuracy and fresh-transfer success 1.0 with zero harm, while the random-diagnostic control reached 0.833 on both measures. Direct self-edit had accuracy 0, false-broad rate 1.0 and harmful-regression rate 0.354. The independent decision- and execution-layer checks agreed with the primary scorer.

None of the seven frozen decision terminals fired: the execution receipt records `terminal_under_frozen_rules = NO_TERMINAL_UNDER_FROZEN_RULES`, and that is the registered outcome rather than an absence of one. The panel registered two expected diagnostics for every hypothesis while the subject policy observed one discriminator per bounded session; its responsibility gate therefore kept every case unresolved until all modeled discriminators were observed. The frozen panel and rules are retained unchanged. Within this panel H3's cannot-check blocking and H4's history-blind re-admission check passed, whereas H2 was not confirmed because the subject remained at the safety floor. These are bounded secondary observations only: the receipt sets `grants_scientific_authority=false`, H1 has no registered terminal, and no general superiority or peer-review-readiness claim follows.

Table 1: Hidden-cause attribution confusion matrix from the archived glm-5.2 diagnostic run over a fixed constructed suite. Accuracy is 21/24; the nominal Wilson score interval [0.690, 0.957] treats the 24 cases as Bernoulli units and is not a population confidence interval. Three residual errors are retained and are not relabeled as successes. This is not a protected fresh-transfer or H1 result.

gold / attr.	Retr	Rout	Impl	Env	Eval	Repr	Meas	Meth
Retr	2	0	0	0	0	1	0	0
Rout	0	3	0	0	0	0	0	0
Impl	0	0	3	0	0	0	0	0
Env	0	0	1	2	0	0	0	0
Eval	0	0	0	0	3	0	0	0
Repr	0	0	0	0	0	2	0	1
Meas	0	0	0	0	0	0	3	0
Meth	0	0	0	0	0	0	0	3

Residual errors: P5-HC-002: gold RETRIEVAL_MISS → REPRESENTATION_GAP (MEDIUM) P5-HC-012: gold ENVIRONMENT_DEPENDENCY_TOOL_FAILURE → IMPLEMENTATION_BUG (HIGH) P5-HC-018: gold REPRESENTATION_GAP → METHOD_BASIS_GAP (HIGH).

Table 2: Retained residual attribution errors from the archived glm-5.2 diagnostic run. These three cases remain incorrect; they are not successes.

case	gold cause	attributed cause	confidence
P5-HC-002	retrieval miss	representation gap	medium
P5-HC-012	environment dependency or tool failure	implementation bug	high
P5-HC-018	representation gap	method basis gap	high

Post-outcome public-suite instrument diagnosis

A separate NR-01 instrument study replays the unchanged 24-case public hidden-cause suite. Its V1-control arm exactly reproduces 21/24 with standard macro-F1 0.872619, including errors P5-HC-002, P5-HC-012 and P5-HC-018. A staged-extraction-plus-deterministic-diagnosis instrument returns 24/24 with standard macro-F1 1.0 on that same suite. This is an instrument-stage positive, not evidence that a model acquired new attribution capability: the requested model was GLM-5.2, the recorded served model was GLM-5.3, and the control comparison pins the instrument rather than a model version.

The protocol, rows, report and revival receipt first enter repository history together, so independent pre-outcome registration chronology is CannotCheck. The suite was already public and frozen, the rows carry no signed provider or independent-custody identity, and no blind, fresh or source-disjoint transfer was run. The admissible disposition is therefore `POST_OUTCOME_PUBLIC_SUITE_INSTRUMENT_DIAGNOSIS_ONLY`. The historical 21/24 diagnostic remains immutable; H1–H4, protected freshness, comparative performance and superiority remain CANNOT_CHECK.

Replay-versus-fresh scatter, longitudinal specialist regression, the improvement/integrity frontier, failure recurrence, cost to protected improvement, the matched baseline and ablation comparison, and the campaign record of harmful and null interventions all have no admissible rows in this archive. Each of the seven tables below therefore names the measurement it would carry and leaves its result undetermined; no number in them is imputed from the diagnostic accuracy above.

Table 3: Replay-vs-fresh scatter: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no campaign-level replay/fresh score pairs. This table will be populated when a live protected-transfer campaign produces matched replay and fresh-task measurements.

Round	Replay score	Fresh score
`CANNOT_CHECK` — no admissible data in archive		

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

Table 4: Longitudinal specialist regression: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no longitudinal specialist-designation data. This table will be populated when a live campaign produces round-by-round specialist scores with regression annotations.

Round	Specialist	Prior score	Current score
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

11 Limitations and threats to validity

This revision establishes an implemented governance path, a local hostile falsifier suite, a frozen prospective design, and one diagnostic attribution archive. It does not establish transferable protected improvement, reduced harmful transfer, or an integrity advantage against matched self-edit or self-evolution baselines.

The theory is wider than the evidence. The revision-factorization, Bayes-risk and discriminator-cover results apply to arbitrary finite decision alphabets and, for the cover theorem, finite latent fibres with deterministic protected test outcomes. The stochastic transcript-law theorem characterizes a fixed adaptive policy. The state-indexed frontier additionally enumerates optimal finite-horizon policies, but only for finite observable Markov-sufficient laboratory state, finite legal actions, known kernels and declared losses. It does not validate those kernels, identify unknown kernels from data, solve the unrestricted infinite-horizon problem, or supply an unknown-kernel sample-complexity bound. The results do not show that the eight revision classes are complete, that the true latent situation is in the registered fibre, or that the required protected distributions are identifiable from finite samples. A registered cover proves structural separability under its model, not empirical power, calibration, transfer or beneficial revision.

What is not claimed as novel. This is not the first self-improving agent, the first evolutionary program-search system, the first candidate archive, the first issue-centric self-improvement state, or the first failure-driven adaptation loop. Meta-agent design search, archive-based self-modification, persistent issue state, multi-hypothesis attribution, counterfactual repair, source-level self-rewriting with replay validation, pre-commit gating, evaluator locking and anytime-valid acceptance are all prior work; they are discussed and cited in Section 8, and each is absorbed rather than claimed. What is left as a candidate difference is the composition: the ordered, non-compensatory sequence from persistent issue state through causal discrimination, invention readiness, isolated execution, replay, independent protected fresh transfer and protected assurance, with promotion authority

Table 5: Improvement-integrity frontier: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no frontier-pair data. This table will be populated when a live campaign produces matched improvement and integrity scores across rounds.

Round	Improvement score	Integrity score
`CANNOT_CHECK` — no admissible data in archive		

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

Table 6: Failure recurrence: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no recurrence annotations. This table will be populated when a live campaign records which previously resolved failures recur across rounds.

Original failure	Round resolved	Round recurred	Outcome
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

structurally outside the evaluated system. That is a claim about the safety and reliability of a development process, and architecture alone does not establish it.

The diagnostic archive does not test the primary hypothesis. The archive scores 21/24 on hidden-cause labels. That quantity is a single-model, single-run descriptive accuracy. Its nominal Wilson interval treats 24 fixed constructed cases as Bernoulli units; without a probability sampling frame or a licensed dependence model it is not a population confidence interval. It does not measure protected fresh-task improvement, harmful transfer, or false protected acceptance. The three residual errors remain errors.

The 24-case suite is not an execution-frozen campaign. The suite covers eight families, but evaluator hashes and several fresh-task hashes remain placeholders. Motivating, replay and fresh split identities, provider and model revisions, and the protected evaluator epoch are all unbound in both frozen protocol revisions.

The wide source universe is not bound. An outcome-blind successor fixes 16 candidate roots across two waves, with four metadata providers, four artifact modalities, four nominal domains and two roots per domain in each wave. Its 13/13 conformance checks and 16/16 live identity checks close metadata structure only. Eight roots lack exact historical Crossref bytes; content-class rights and case eligibility are unresolved for all sixteen; and no project/issue case, protected revision label or outcome has been opened. The 768-cluster successor therefore remains unexecuted.

Public bug corpora widen development coverage but do not create protected evidence. SWE-bench Multilingual, Defects4J, BEARS, GitBug-Java, Bugs.jar, BugSwarm, BugsInPy, ManyBugs, QuixBugs and SWE-bench Verified now have metadata identities, but framework or repository licences do not automatically license every upstream project, issue, patch, test, log, dependency or container byte. None supplies protected eight-class revision responsibility or protected fresh transfer. Payload access therefore remains conditional on a per-project and per-content-class rights/eligibility census performed without outcome-based source selection.

Table 7: Cost to validated improvement: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no cost-to-improvement annotations. This table will be populated when a live campaign records the resource cost (e.g., API calls, rounds, wall time) required to reach validated improvement.

Improvement	Validation	Cost metric	Value
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

Table 8: Baseline and ablation transfer/integrity results: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no baseline/ablation arm data. This table will be populated when a live campaign produces matched baseline-vs-ablation transfer and integrity measurements.

Condition	Transfer score	Integrity score	Δ vs. baseline
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

No matched baselines or ablations were executed. The six comparator identities are source-bound, but no matched comparison has been executed. SWE-agent, MOSS, Darwin Gödel Machine, ADIAS, Double Ratchet's metric-only arm and ScienceClaw each have an exact paper/repository revision and native entrypoint. A generic refined adapter contract is now total on its declared synthetic domain. C1 SWE-agent now has an outcome-blind native parser/write-isolation and partial resource binding but still has 12 blocking fields. C2 MOSS separately has a native parser, closed fallback policy, wallclock and dependency locks but retains 14 blockers; its authoritative source tree omits the benchmark runtime required by its native trial runner. C3 DGM has a closed patch parser, fallback policy and outer wallclock but retains 15 blockers, including a resolved dependency lock: its 23 declarations have zero exact pins. C4 ADIAS has a closed report parser, fallback/search policy and outer wallclock but retains 15 blockers, including dependency/image identity, task rights and rootless isolation. C5 Double Ratchet has an isolated write surface, parser, resource envelope and dependency lock but retains 12 blockers because the official loop regenerates solver outputs and reuses its development-locked metric every round. C6 ScienceClaw has a strict draft-shape parser and outer wallclock but retains 16 blockers: no native revision selector, no close-all fallback switch, no locked 112-declaration/lazy-dependency environment, and no complete resource, rights or custody binding. None has a protected panel, complete matched resource manifest or result-bearing archive on this subject. Identity closure therefore narrows the blocker from which systems to whether their native interfaces can be translated lawfully and executed under equal exposure without protected feedback. It supplies no effect estimate, and the matched baseline and ablation comparison remains undetermined rather than null.

Adapter conformance is not comparator performance. The six-arm V2 adapter audit proves a formal interface boundary and passes 90/90 synthetic contract cases; V3 extends conformance to 231/231 declared fictional cases. Neither uses a native output example or outcome. All six raw same-visible-symptom fibres are mixed, so zero raw native statuses license a singleton class. V3's 40 singleton case records occupy 20 supported constant fibres only after a host-validated class certificate and exclusive-front write proof are supplied; the other 191 cases remain `UNRESOLVED`.

Table 9: Campaign harmful/null interventions: no admissible rows in the archived glm-5.2 attribution run. The archive contains a single-model hidden-cause diagnostic over 24 cases and no intervention-level outcome annotations. This table will be populated when a live campaign records which interventions were harmful, null, or beneficial relative to the incumbent.

Intervention	Outcome	Effect size	Recovery
`CANNOT_CHECK` — no admissible data in archive			

Empirical authority: DESCRIPTIVE_ONLY (diagnostic attribution). Not a protected fresh-transfer or H1 result.

These records show only that a proposed action can be typed when the missing identifying information is supplied. They do not show that the action is correct, minimal, preserving, transferable, or safe. ScienceClaw has no native singleton support, Double Ratchet conditionally supports only evaluator repair, and the V3 ledger's 108 blocking fields are preserved as its historical result. The C1 and C5 V4 packets each resolve six fields, while C2, C3 and C4 resolve respectively four, three and three fields, while C6 resolves two. This leaves 12 each for C1/C5 plus 14 for C2 plus 15 each for C3/C4 plus 16 for C6, or 84 current blocker instances. Zero of six arms has a complete matched resource, rights, arm-native adapter and protected-scorer manifest.

Root-cause compression is not blocker closure. The panel-wide V5 registry proves that the 84 current blocker instances are an exact partition into 14 equivalence classes and five evidence-producing work packages. This permits shared acquisition where the same evidence is logically sufficient. V5 itself does not permit a schema to fan out as evidence and its arm-field delta is zero. V6 subsequently provides one substantive rights-cleared visible-case packet and exact per-arm acceptance receipts, closing its targeted 12 fields and reducing the census to 72 blockers. It does not by itself close the six native task environments or the 18-field independent-custody root. V7 later closes only C1's environment with actual offline setup/configuration bytes, reducing the census to 71 blockers and the environment root to five arm-specific bundles. Moreover, the wallclock bindings are not matched: C1's 3,600-second run and 30-second grace differ from C2–C6's 21,600-second run and 120-second grace, and C5 has an extra 10-second candidate subprocess limit. Equal nominal field status is therefore insufficient for equal effective exposure.

The C3 V8 candidate-safe seed does not close its environment. Its unchanged native parent-selection path is outcome-indexed: a fresh singleton `initial` parent is skipped unless prior performance and identifier partitions exist. V9 establishes this boundary only for the exact released DGM commit. It does not prove that outcome-free Darwin Gödel Machine initialization is impossible in future designs. A future source-native identity element can reopen the gate; a locally invented score-free selector is instead a new comparator requiring separate preregistration and matched-panel review.

Fresh source-packet assurance is not uniform. The C4 validator includes the hash of metadata containing a random temporary generation path in its audit receipt. Re-running it changes `AUDIT_RECEIPT_V4.json`, so the frozen C4 packet checksum does not reproduce without a path-invariant repair. The C6 validator requires a cleaned `.source-audit` checkout that is no longer present, so its fresh source-level rerun is CannotCheck. The field-registry, result and parser bytes used for the 42/84 census remain directly hash-verifiable; that narrower assurance must not be expanded into full packet reproducibility.

Common content rights are not native-environment rights. The V6 Lang-1 core is substantive and lawful for its declared candidate-visible task. Its exact acceptance is conditional on every adapter using only that core. C1's image/setup/run configuration is now bound. C2's benchmark companion and native session, C4's domain/tasks/seeds/limits, C5's solver outputs/prompts/membership and C6's profile/skills/tools/fallback closure remain absent. C3 additionally lacks an outcome-free native initial parent; its host/agent split and endpoint bytes alone cannot supply one. Reintroducing bundled native content triggers a new component-rights audit rather than inheriting the common grant.

The released MOSS runtime is incomplete at the pinned identity. The paper-linked commit, tree and release archive are byte-addressed, and the root/OpenClaw licence layers are retained. However, the native trial runner imports and writes under `benchmark/claw-eval`, while the complete non-truncated authoritative tree contains no `benchmark/` path and its setup instructions do not fetch one. Source availability therefore does not establish task-runtime or benchmark rights. This absence is retained as CannotCheck; it is not repaired with invented fixtures, unversioned bytes or a different evaluator.

The C2 successors are lawful components, not an assembled MOSS arm. V11, V12 and V13 close different fields for three distinct authored successor identities. V11's runtime and V12's source-core rights are not inherited by V13; V11 likewise does not inherit V12 or V13, and V12 does not inherit V11 or V13. Adding 8/21, 9/21 and 8/21 states would therefore manufacture an unexecuted fourth identity. V13's scratch probe establishes a complete nine-file image rights/SBOM chain and a constrained container runtime receipt, but that probe is not the missing released MOSS benchmark runtime and does not bind `runtime.container_or_environment` for released MOSS. Released MOSS remains 7/21 bound and 14 blocking, the 55/126 panel census is unchanged, and no performance, H1–H4 or superiority inference follows.

The released DGM runtime is not a frozen execution environment. The source commit, tree, uncompressed Git-archive digest and Apache-2.0 root licence are byte-addressed. That does not bind execution: all 23 Python dependency declarations are unpinned, the Docker base image is mutable, and the candidate source includes outcome/initial paths that must be excluded before access. The released outer-loop timeout is called only after a future has completed, and the source retains an argparse choice concatenation defect and a destructive-untrusted-code warning. These findings are source preflight, not a failed DGM performance observation; native execution remains CannotCheck.

The released ADIAS runtime is not a rights-closed isolated execution environment. The source commit, tree, prefixed Git-archive digest, licence and 1,578-file manifest are byte-addressed. The root source licence is CC-BY-NC-SA-4.0 and does not itself close the exact intended-use rights; bundled task data expose no component licence/NOTICE files under `data/`. Four VCS declarations are unpinned, the base image is mutable, native containers use host networking with provider keys, and a root agent defeats the source's `chmod 0` domain-hiding convention as a security boundary. Same-run final testing is not an external protected scorer. These are source-preflight defects, not failed ADIAS performance observations; native execution remains CannotCheck.

The released Double Ratchet metric loop is not a protected scorer. The source, entry-point, licence, 113-blob tree, prefixed Git archive and 46-entry dependency lock are byte-addressed, and V4 adds a results-only write surface and resource envelope. Yet the official loop regenerates hosted solver outputs and evaluates, persists and prints `eval_locked` each round. The tree contains no frozen split/result payload, and its MBPP/report-generation surface is not a P5 dossier/eight-class task adapter. A development-only surrogate may remain inside the loop, but the final fresh panel must be external and one-shot. These are interface/custody findings, not a failed Double Ratchet performance observation; native P5 execution remains CannotCheck.

The residual-error successors retain distinct execution status. The three factorial discriminators were derived from retained errors and are therefore new studies, not confirmatory rescoring of the original 24 cases. P5-RD-01 and P5-RD-03 have no observations. P5-RD-02 V1 remains CANNOT_CHECK with an adverse-cell modifier; V2 records one preregistered public-development known-fix cluster with a within-cluster IMPLEMENTATION_MAIN_EFFECT; and V3 is a post-outcome audit-only archival replay. V2 does not rewrite V1, V3 does not recover V2's overwritten checkout streams, and none changes P5-HC-012 or supplies protected freshness. A main effect, interaction, joint null, adverse terminal, or persistent observational equivalence remains attached to its own study identity; none may be collapsed into a post-hoc success.

The 24/24 V2 attribution result is an instrument-stage diagnosis. The V2 study reuses the same public 24-case suite and changes the attribution instrument from free-form causal rederivation to staged evidence extraction plus deterministic diagnosis. Its control reproduces 21/24 and its treatment reaches 24/24, but protocol and outcomes first appear in the same commit; independent preregistration chronology is therefore CannotCheck. The requested GLM-5.2 identity also differs from the recorded GLM-5.3 served identity. Without signed provider custody, a blind fresh suite, source-disjoint transfer or a matched protected panel, this result cannot establish model improvement, general attribution accuracy, H1–H4 or superiority.

The frozen revision-level panel exposes an identification mismatch. All bound execution identities verified and the independent scorers agreed, but FULL_T7 observed one discriminator while each hypothesis expectation map required two. The resulting 0.125 accuracy and fresh-transfer rate are therefore a measured property of the frozen one-probe mechanism on this panel, not a software failure and not a result that may be erased by changing the panel after outcome access. The generic-diagnosis and random controls outperform the subject on this constructed panel. A successor may test completable expectation sets and explicit preservation conflicts, but it must be separately frozen; it cannot retroactively turn this no-terminal result into support for H1 or general superiority.

No live development cycle is bound. The live-trial packet still records an unbound corpus revision. Provider and protected-verifier credentials were unset for this revision, so no live campaign was started. A previously advertised report of a flawless run is not used: its artifact identity and repository identity diverged before integration.

Statistical threats. The diagnostic contains 24 fixed constructed cases from one model run, not 24 independent campaigns or a probability sample. P5-RD-02 contains one source/bug/fix cluster; its environments, cells and phase streams are repeated within-cluster measurements, not independent replicates. Neither archive can support the predeclared five-percentage-point fresh-improvement target or the two-percentage-point harm guard. There is no paired cluster-level bootstrap against a baseline, no multiplicity-adjusted secondary family, and no tail or family regression from a campaign. The macro-F1 reported in the original sidecar assumed flawless precision and is not promoted.

Contamination and evaluator custody. The diagnostic run used locally visible gold labels to score attributions after the fact. That is admissible for an offline label-recovery diagnostic. It is not admissible as protected-evaluator evidence: the candidate-facing study still requires labels, fresh payloads and scoring rubrics to remain outside challenger custody.

12 Data and code availability

The implementation repository contains the development controller, the protected assurance path, the local hostile falsifier suite, the hidden-cause case suite, the diagnostic attribution archive and the table generator. Every artifact named in the narrative is listed here with its path, so that no repository location has to be carried in the prose. Source, protocols and evidence are version-controlled in the `SzeChunYiu/ORION` repository, under `papers/orion-15-self-orion/`; persistent DOI assignment remains pending publication.

Preregistration

The original protected-campaign design was frozen before that campaign accessed any outcome, under the identifier `P5.hidden-cause-fresh-transfer.v1`. Its recorded state is a design freeze, not an execution freeze. A reader checking the prospective margins in Section 9 should compare them against that artifact. This timing claim does not extend to every later successor: P5-RD-02 has separate V1/V2 development identities, and V3 was frozen after V2 outcomes were known for an audit-only archival replay.

Frozen design, case structure and acceptance rules

The design freeze is `protocol/PROTOCOL_V1.json`; the hidden-cause case structure is `protocol/HIDDEN_CAUSE_CASE_SCHEMA_V1.json`; replay/freshness definitions and harm accounting are `protocol/FRESH_TRANSFER_POLICY_V1.md`; and the staged acceptance rules and their successor revision are `protocol/PROTOCOL_V2.json` and `protocol/STAGED_ACCEPTANCE_POLICY_V2.md`. The protected-suite freeze procedure is `protocol/PROTECTED_SUITE_FREEZE_V1.md`, and the retention chain for negative history is `protocol/P5_NEGATIVE_HISTORY_CHAIN_V1.json`. The revision-level successor benchmark of Section 9 is frozen as `protocol/SELF_ORION_V3_REVISION_LEVEL_PROTOCOL_V1.json`; the additive metadata-only correction that binds MDA v3 structurally while retaining the unreleased-solver blocker is `protocol/SELF_ORION_V3_MDA_SOURCE_BINDING_AMENDMENT_2026-08-23.json`. The external transfer benchmark used as a reference point is pinned to `Gen-Verse/PAST-Bench@f8223517ae7491e776b69793d9f11e9d074ab42e`; any use of it still requires a frozen task and split manifest.

The additive wide design is `protocol/P5_WIDE_REVISION_LEVEL_SUCCESSOR_V1.json`; its three error-derived identification studies are `protocol/P5_RESIDUAL_DISCRIMINATOR_SUCCESSORS_V1.json`. Those protocol files record their outcome-blind state at their own design-origin timestamps; they are not a current claim that every descendant study remains unexecuted. The wide campaign, P5-RD-01 and P5-RD-03 remain unexecuted. P5-RD-02's later public-development V1/V2 observations and post-outcome V3 archival audit are listed below. None modifies the original preregistration, corrects a diagnostic error, or turns public replay into protected confirmatory evidence. The provider- and modality-diverse candidate frame is `development/pr` ovider-diverse-metadata-census-2026-08-23/SOURCE_CENSUS_V1.json, with protocol development/provider-diverse-metadata-census-2026-08-23/CENSUS_PROTOCOL_V1.json and receipt development/provider-diverse-metadata-census-2026-08-23/VERIFICATION_RECEIPT_V1.json. These artifacts contain no project/issue cases, labels, outputs or outcomes; their rights and eligibility terminals remain fail-closed. The outcome-blind six-arm identity and public-panel preflight is archived under development/p5-six-arm-comparator-panel-2026-08-23/. Its protocol, normalized identity ledger, public-source rights matrix, retained negative-result ledger, primary-source research sidecars and checksum manifest distinguish semantic/source identity from execution readiness and record each official-versus-reimplementation status, licence byte, entry-point, information/action/cost envelope, native error terminal and missing adapter. No comparator performance, dataset row or protected outcome is present. The terminal-preserving adapter successor is archived under development/p5-six-arm-terminal-adapters-v2-2026-08-23/. Its prospective protocol and six same-visible-symptom fixtures were frozen before any native output example; the per-arm ledgers, resource/configuration audit, factorization theorem, deterministic conformance program, 90-case receipt, recursive negative-result ledger and SHA256SUMS retain native failures and unsupported or mixed fibres as UNRESOLVED. The archive contains no comparator execution, native output example, public or protected outcome, or performance estimate. It leaves zero of six arms confirmatory-ready and preserves H1–H4 and protected freshness as CANNOT_CHECK. The stricter outcome-free refinement is archived under development/p5-six-arm-adapter-refinement-v3-2026-08-23/. Its frozen certificate schema, action/write-surface schema, eight-class front registry, native-terminal rules and deterministic validator cover exactly 231 declared synthetic cases. The receipt records 40 typed case records in 20 supported constant fibres and 191 UNRESOLVED, with zero raw-native licences and zero conformance failures. The separate execution ledger preserves 108 exact blocker instances, 18 per arm, and zero of six execution-ready. Its SHA256SUMS also pins the unchanged V2 predecessor manifest. This is contract conformance only: no comparator, native example, public/protected outcome or performance table was opened.

The outcome-blind C1 SWE-agent execution-binding successor is archived under development/p5-swe-agent-execution-binding-v4-2026-08-23/. Its field registry, pinned source/rights manifest, native parser and output schema, fail-closed write-surface wrapper, resource/dependency lock, candidate-case and custody schemas, adverse-result ledger, audit receipt and SHA256SUMS bind 9/21 C1 fields and preserve 12 blockers. The exact terminal is P5_C1_V4_SWE_AGENT_NATIVE_PARSER_AND_ISOLATION_POLICY_BOUND__TWELVE_C1_FIELDS_BLOCKING__ZERO_OF_SIX_PANEL_READY__PERFORMANCE_AND_SUPERIORITY_CANNOT_CHECK. The packet copies only an outcome-free 101-byte upstream smoke fixture and no substantive P5 case. No comparator/model, protected scorer, gold patch, protected-test value, performance outcome or superiority result was opened.

The distinct outcome-blind C2 MOSS execution-binding successor is archived under development/p5-moss-execution-binding-v4-2026-08-23/. Its field registry, exact source/rights manifest, native parser and terminal rules, fail-closed runner, host and OpenClaw locks, smoke receipt, adverse-result ledger and 70-check audit bind 7/21 C2 fields and preserve 14 blockers. The exact terminal is P5_C2_V4_MOSS_NATIVE_PARSER_DEPENDENCY_LOCKS_FALLBACK_AND_WALLCLOCK_BOUND__FOURTEEN_C2_FIELDS_BLOCKING__RELEASED_BENCHMARK_RUNTIME_ABSENT__ZERO_OF_SIX_PANEL_READY__PERFORMANCE_AND_SUPERIORITY_CANNOT_CHECK. The authoritative commit/tree/archive are pinned, but the tree has zero benchmark/ paths although the native runner requires benchmark/claw-eval. No model, benchmark, protected case, scorer, performance outcome or superiority result was executed or opened.

The distinct outcome-blind C3 DGM execution-binding successor is archived under development/p5-dgm-execution-binding-v4-2026-08-23/. Its exact source/tree/uncompressed-Git-archive identity, source-rights manifest, closed native parser, terminal rules, empty fallback policy, fail-closed runner, resource registry, candidate-case and custody schemas, negative ledger and 134-check audit bind 6/21 C3 fields and preserve 15 blockers. Its manifest contains 23 entries. The exact terminal is `P5_C3_V4_DGM_SOURCE_NATIVE_PARSER_EMPTY_FALLBACK_AND_OUTER_WALLCLOCK_BOUND__UNPINNED_DEPENDENCIES_NOT_MISLABELED_AS_LOCK__FIFTEEN_C3_FIELDS_BLOCKING__ZERO_OF_SIX_PANEL_READY__PERFORMANCE_AND_SUPERIORITY_CANNOT_CHECK`. The dependency declarations have zero exact pins and no lockfile. The packet opens no released outcome payload and executes no DGM, model, task, benchmark, Docker container, protected case, score, performance outcome or superiority contrast.

The distinct outcome-blind C4 ADIAS execution-binding successor is archived under `development/p5-adias-execution-binding-v4-2026-08-23/`. Its exact source/tree/prefixed-Git-archive identity, 1,578-file source manifest, source-rights record, closed native parser and terminal rules, empty fallback policy, fail-closed runner, resource and dependency-declaration registries, candidate-case/custody schemas, negative ledger and 35-check audit bind 6/21 C4 fields and preserve 15 blockers. Its manifest contains 23 entries. The exact terminal is `P5_C4_V4_ADIAS_SOURCE_TREE_NATIVE_PARSER_EMPTY_FALLBACK_AND_WALLCLOCK_BOUND__FIFTEEN_C4_FIELDS_BLOCKING__DEPENDENCY_LOCK_TASK_RIGHTS_ISOLATION_AND_CUSTODY_UNBOUND__ZERO_OF_SIX_PANEL_READY__PERFORMANCE_AND_SUPERIORITY_CANNOT_CHECK`. Four VCS declarations remain unpinned and task rights/isolation/custody remain unbound. The packet executes no ADIAS model, domain, benchmark, task episode, protected case, score, performance outcome or superiority contrast.

The distinct outcome-blind C5 Double Ratchet metric-only execution-binding successor is archived under `development/p5-double-ratchet-execution-binding-v4-2026-08-23/`. Its exact source/tree/prefixed-Git-archive, source-rights and 113-blob metadata manifests, native parser/terminal rules, isolated write surface, empty fallback policy, resource envelope, 46-entry uv lock, custody schemas, negative ledger and 351-check audit bind 9/21 C5 fields and preserve 12 blockers. Its manifest contains 24 entries. The exact terminal is `P5_C5_V4_DOUBLE_RATCHET_SOURCE_PARSER_ISOLATION_FALLBACK_WALLCLOCK_COMPUTE_AND_DEPENDENCY_LOCK_BOUND__TWELVE_C5_FIELDS_BLOCKING__OFFICIAL_RUNNER_REGENERATES_SOLVER_OUTPUTS_AND_REPORTS_DEVELOPMENT_LOCKED_EACH_ROUND__ZERO_OF_SIX_PANEL_READY__PERFORMANCE_AND_SUPERIORITY_CANNOT_CHECK`. The official loop regenerates hosted solver outputs and exposes its development-locked metric every round; no P5-native task adapter or external one-shot panel is bound. The packet executes no model, solver-output set, benchmark, P5 case, protected score, performance outcome or superiority contrast.

The distinct outcome-blind C6 ScienceClaw execution-binding successor is archived under `development/p5-scienceclaw-execution-binding-v4-2026-08-23/`. Its exact source/tree identity, source-rights record, strict dry-run-draft parser, terminal rules, outer wallclock, case/custody schemas and negative ledger bind 5/21 C6 fields and preserve 16 blockers. No native P5 selector or close-all fallback switch exists, and its thirteen prior-result-prefix files were hashed/censused without decoding. The packet executes no investigation, model, tool, service, benchmark, prior-result payload or protected outcome. Fresh source-level validator assurance is CannotCheck after cleanup because the required `.source-audit` checkout is absent.

The panel-wide root-cause packet is archived under `development/p5-panelwide-root-closure-v5-2026-08-23/`. Its protocol, cross-arm equivalence registry, result, recursive negative ledger, fail-closed metadata gate, 74-check validator and 12-entry checksum manifest recompute 42/126 bound and 84/126 blocking with zero of six ready. Fourteen equivalence classes are partitioned into five root work packages, reducing the coordination queue from 84 entries to five while creating zero arm-field bindings. The packet also retains the unmatched wallclock envelopes and the C4/C6 assurance defects. It executes no arm, model, benchmark or protected scorer and opens no protected outcome; H1–H4, performance and superiority remain CANNOT_CHECK.

The first root-closure successor is archived under `development/p5-common-visible-case-rights-v6-2026-08-23/`. Its exact Lang-1 source manifest, provenance, rights manifest, six-arm acceptance receipt, recursive ledger, 741-check validator and 22-entry checksum manifest bind 12 shared fields and recompute 54/126 bound, 72/126 blocking and zero of six ready. The public source bodies are candidate-visible development content; no arm, model, protected scorer or outcome is executed. Any future adapter that adds content outside the frozen common core must reopen the corresponding rights field.

The byte-level environment successor is archived under `development/p5-native-task-environment-fanout-v7-2026-08-23/`. It binds C1's exact offline setup and effective configuration, records five distinct arm-specific residual bundles, and recomputes 55/126 bound, 71/126 blocking and zero of six ready. Its validator reports one environment bound, five blocking, 67 referenced bytes verified and 25 packet files hashed. It executes no arm, model, benchmark, protected scorer or outcome.

The C3 candidate-safe environment and its outcome-free initialization stop are archived under `development/p5-c3-native-environment-v8-2026-08-23/` and `development/p5-c3-outcome-free-successor-v9-2026-08-23/`. V8 binds the 69-member seed, six unchanged Lang-1 members, eight control gates and the exact native-initialization failure while opening none of the 1,595 excluded payloads. V9 retains the preregistration, exact-byte/AST receipt and impossibility witness without materializing an initializer. Both leave `runtime.task_environment` blocking; no DGM, model, benchmark, scorer or outcome is executed, and the prohibited C4 validator is not rerun.

The three non-aggregable C2 lawful-successor packets are archived under `development/p5-c2-lawful-native-byte-successor-v11-2026-08-24/`, `development/p5-c2-lawful-native-byte-successor-v12-2026-08-24/` and `development/p5-c2-lawful-native-byte-successor-v13-2026-08-24/`. V11 binds only `runtime.task_environment` for a distinct 8/21 successor; V12 binds only candidate-visible bytes and task-content rights for a distinct 9/21 successor; V13 binds only container/generated-artifact rights for a distinct 8/21 successor. V13 contains the complete nine-file SPDX 2.3 SBOM, rights map, licence bundle, generated-artifact authority, exported-layer manifest, constrained runtime receipt, image archive and remote-environment destruction receipt. The additive read-only integration audit is `development/p5-c2-v11-v13-manuscript-integration-audit-2026-08-24/`. Its byte-identical paper-bound copy is `evidence/P5_C2_V11_V13_INTEGRATION_AUDIT.json`. Each packet's validator and `SHA256SUMS` pass without pytest or repository CI. The three states are not unioned; released MOSS remains 7/21 bound and 14 blocking, 0/6 arms are ready, and H1–H4, performance and superiority remain CANNOT_CHECK.

The mapping from every result-bearing sentence in this manuscript to the archived artifact that supports it is `evidence/CLAIM_LEDGER_V1.json`, with a human-readable view at `evidence/CLAIM_LEDGER_V1.md`. The local hostile falsifier suite is `evidence/FALSIFIER_V1.md`; the protected hidden-cause suite is `evidence/hidden-cause-suite/PROTECTED_SUITE_V1.json`; and the nearest-work dispositions that Section 8 discusses in prose are recorded per mechanism in `evidence/TABLE_P5_1_NEAREST_WORK.json`.

Revision-level confirmatory archive

The executed 96-case panel is bound by `protocol/SELF\_ORION\_V3\_REVISION\_LEVEL\_PROTOCOL\_V1.json`. Its human-readable result is `evidence/revision-level-v3/CONFIRMATORY\_EXECUTION\_RESULT\_2026-08-24.md`, and the machine-readable receipt with per-policy metrics and independent cross-checks is `research/self-orion-v3/confirmatory/CONFIRMATORY\_EXECUTION\_RECEIPT\_2026-08-24.json`. The archive records `NO_TERMINAL_UNDER_FROZEN_RULES` and `grants_scientific_authority=false`; it must not be represented as a positive H1 or superiority result.

Diagnostic attribution archive

The raw per-case records are `evidence/glm-5.2-attribution/results.jsonl`, the checked sidecar report is `evidence/glm-5.2-attribution/report.json`, and the reproduction receipt is `evidence/glm-5.2-attribution/REPRODUCTION_RECEIPT.json`. The generated table data are under `evidence/tables/` and their rendered forms under `manuscript/tables/`. Architecture diagrams and pseudocode for the controller, and the historical technical companions this design inherits from, are retained under `research/technical-companions/`.

P5-RD-02 public-development factorial and archival audit

The three distinct Defects4J identities, their preregistrations, frozen-input receipts, runners, raw phase streams and results are under `development/p5-rd02-defects4j-public-factorial-2026-08-23/`. The bounded scientific interpretation and immutable adverse-history map are `SCIENTIFIC_REPORT.md` and `NEGATIVE_RESULT_LEDGER.md`; the rights, authority and stream-provenance audit is `AUDIT_RECEIPT.json`, with every retained artifact bound by `SHA256SUMS`. The V3 result digest is `e743ebb77a31c4c4606f7eded8045a60c78fa2f251c13908051156cfd4a1fb6e`. These artifacts support one local public known-fix replay and an archival audit only. They contain no protected/fresh task and grant no H1–H4 authority.

13 Reproducibility

Regeneration of every archived result, without executing a live system, is

```
make paper05-results
```

which runs `python -m orion.study.p5.tables`. The command recomputes 21/24 from the raw records, writes the confusion matrix and the residual-error ledger, and writes stubs for the seven tables that have no admissible rows. A rewrite of the archive that hides the three residual errors is refused with an error exit; a missing archive is reported as undetermined under its own distinct exit code rather than as a build failure, so that an absent measurement cannot be read as a failed build or as a negative result. Exact commands are in `papers/orion-15-self-orion/REPRODUCE.md`. Continuous integration regenerates these artifacts on a clean checkout and requires every committed file to be byte-identical to what the command produces.

Compute and conservative-abstention cost

The diagnostic archive records 14 190 tokens and a mean latency of ≈13 s per case, for attribution only. Cost and time to *protected validated improvement* remain undetermined. No campaign-level abstention/cost trade-off is reported.

Licensing

This revision does not add a repository-level license. Redistribution terms therefore remain undetermined from the repository state until they are declared explicitly.

14 Ethics and safety

This work studies governed method evolution for research software. The frozen local falsifier and the diagnostic attribution suite use constructed hidden-cause cases. They do not involve human-subject records or personally identifiable information. No live provider campaign was executed in this revision.

The controller has no self-merge primitive, and its strongest internal terminal is a host-only promotion recommendation. Isolated self-certification ablations, if ever executed, are forbidden on the production authority path. Negative, null, blocked and undetermined outcomes are retained rather than dropped.

15 Conclusion

Method evolution is treated here as a minimal-revision and authority problem rather than as an invitation for an agent to edit itself. The theory identifies the general obstruction: if one visible interface fibre contains situations requiring different minimal fronts, no downstream algorithm can select exactly; its best attainable loss is the within-fibre Bayes risk. A finite intervention programme resolves the obstruction exactly when it covers every cross-decision pair, and no sound-and-complete self-promotion rule exists when protected authority differs inside an internally identical transcript. For stochastic adaptive programmes the same boundary is distributional: exact decoding requires mutually singular decision-class transcript laws, and overlapping laws retain transcript-conditioned Bayes impurity. Failure episodes therefore remain evidence, recurrence remains distinct from cause, issues persist across discriminators and interventions, invention is gated by search and attribution, challengers execute in isolation, replay is separated from fresh transfer, protected assurance sits outside candidate write authority, and final promotion remains in host custody. The successor formalism sharpens the same boundary: an anomaly is not itself permission to revise, an unresolved responsibility is not a licence to choose among incompatible repairs, an interface defect is not evidence that the method basis should be rewritten, more expected computational value is not scientific authority, a validity region is not a truth certificate, and repeated social reports are not independent evidence merely because they came from different processes.

The local hostile programme demonstrates these registered governance semantics and has already exposed authority failures in earlier readiness designs. The diagnostic archive scores 21/24 on hidden-cause labels and retains three residual errors; that descriptive result supports none of the paper's hypotheses. Those errors now open three separately identified factorial discriminator studies. P5-RD-01 and P5-RD-03 remain unexecuted; P5-RD-02 has only one public known-fix development cluster and a post-outcome archival audit, with no protected/fresh or H1–H4 authority. The protected-custody revision-level successor was frozen and executed on 96 constructed cases; the fully composed subject remained at the safety floor and no registered terminal fired. The wider 768-cluster design remains specified but unexecuted. Neither record supplies a positive protected improvement result. Architecture readiness and analytic separability do not imply that candidate causes are scientifically correct, that the development policy chooses valuable work, that repairs transfer, or that a protected evaluator is sufficient against unknown gaming strategies.

The six-arm terminal-adapter audit makes the interface obstruction concrete. Each raw native status has a mixed same-visible-symptom fibre, so none licenses a singleton revision class. The V2 90/90 receipt and its stricter V3 successor verify only synthetic contract behavior. V3 is total on 231 declared fictional cases: 40 records in 20 supported constant fibres emit a typed proposed action, while 191 retain `UNRESOLVED`; it has no exception, expectation, vocabulary or native-terminal-retention failure. It is not a performance result. ScienceClaw remains native-unsupported, Double Ratchet remains evaluator-only, exact preflight retains 108 execution-blocker instances, zero of six arms is confirmatory-ready, and H1–H4 remain CANNOT_CHECK.

The C1 V4 successor does not weaken this boundary. It converts six SWE-agent blockers into outcome-blind byte-level parser, isolation and resource bindings, so C1 now has 12 blockers. The separate C2 MOSS V4 binds four fields but retains 14 and discovers that the authoritative source tree omits the native `benchmark/claw-eval` runtime. The merged six-arm preflight therefore has 98 blockers. C1's parser still emits `UNRESOLVED` at the responsibility layer, the wrapper refuses execution, C2's full native smoke is CannotCheck, the other four arms are unchanged and panel readiness remains zero of six. This is execution-interface progress, not evidence that a repair is correct or that one system improves over another.

The C3 DGM V4 successor retains the same boundary. It binds three fields beyond the source identities but leaves 15 blocking, including the resolved dependency lock, disposable isolation, exact case bytes, task rights, models, resources and independent custody. Its patch parser always returns `UNRESOLVED` at the responsibility layer. The cumulative six-arm count is therefore 95 rather than 98 blockers, with C4–C6 unchanged and readiness still zero of six. No performance contrast is created by this reduction.

The C4 ADIAS V4 successor makes another interface-only reduction. It binds a native report parser, an empty fallback/search policy and an outer foreground wallclock, while retaining 15 blockers for exact cases, dependency and image identity, bundled-task rights, isolated execution, model/resource bindings and external custody. Its parser also returns `UNRESOLVED`. The cumulative count is therefore 92 blockers, with C5–C6 unchanged and readiness still zero of six. Source inspection and zero-task smoke fixtures create no ADIAS performance contrast.

The C5 Double Ratchet metric-only V4 binds six more execution fields and leaves 12 blocking. Its isolated runner and dependency lock do not solve the scientific leakage boundary: the official loop regenerates hosted solver outputs and evaluates, persists and prints `eval_locked` every round, while no P5-native task adapter, frozen solver-output tree or external one-shot panel is bound. The cumulative count is 86 blockers, with C6 unchanged and readiness still zero of six. Parser/runner conformance creates no performance contrast and licenses no raw native singleton.

The C6 ScienceClaw V4 binds a strict draft-shape parser and outer wallclock, leaving 16 blocking fields. Its released CLI exposes no native revision selector and no switch that closes all model, reasoning, skill-selection and dependency fallbacks; nominal dry-run would execute the investigation before suppressing posting. The parser therefore maps every admissible draft to `UNRESOLVED`. The cumulative count is 84 blockers and readiness remains zero of six. Source hashing and authored shape fixtures create no performance contrast, class licence or independent custody.

The panel-wide V5 root-cause packet is a scheduling and fail-closed evidence contract, not a seventh empirical result. It verifies 42 bound and 84 blocking field instances across the six arms, then partitions the 84 blockers exactly into five evidence work packages. The reduction from 84 queue entries to five packages is coordination compression only: the arm-field delta is zero, readiness is still zero of six, and performance and superiority remain CANNOT_CHECK. In particular, six individually bound wallclock fields do not imply matched exposure because C1's 3,600-second/30-second-grace envelope differs from the 21,600-second/120-second-grace envelope on C2–C6, with an additional 10-second C5 subprocess cap. Nor can local receipt schemas establish independent custody, rights, a built runtime or a native task environment.

V6 then supplies the evidence V5 refused to invent. A complete public Lang-1 candidate core, its Apache-2.0/CC0 rights chain and six exact shared-core acceptance receipts close the candidate-visible-case and task-content-rights fields for every arm. This is 12 genuine field closures, not schema fan-out: 54/126 fields are now bound and 72/126 block. The common-core licence does not cover arm-specific images, companions, solver outputs, profiles, skills or generated artifacts. Consequently the six native task-environment fields and all other V5 roots remain blocking; zero of six is execution-ready and no performance statement follows.

V7 subsequently binds one of those six instances: C1 has exact offline setup bytes and a fixed effective configuration against the unchanged lawful core. This is one genuine field closure, so the current census is 55/126 bound and 71/126 blocking. C2–C6 still lack their arm-specific byte bundles, zero arm is ready, and the C1 acceptance supplies neither matched exposure nor performance authority.

V8 and V9 preserve this census. V8's 69-member C3 seed closes eight materialization/policy checks but fails the load-bearing native-initialization gate because the released DGM requires excluded prior-performance state. V9 shows that no semantics-preserving outcome-free adapter exists for that exact source identity: every apparent repair either restores outcomes, fabricates scores, removes the self-edit action or changes parent selection. This is a source-interface impossibility boundary, not a claim about future DGM designs and not negative performance evidence. C3 remains blocking; H1–H4 remain CANNOT_CHECK.

V11–V13 refine the legal byte-level envelope for three separately authored C2 successors, not for one accumulated comparator. V11 closes the task-environment field in an 8/21 state; V12 closes candidate-visible bytes and task-content rights in a separate 9/21 state; V13 closes scratch-image SBOM, licence and generated-artifact authority in a separate 8/21 state. Their non-aggregation is scientifically load-bearing: none supplies the other states' fields, none modifies released MOSS, and none creates a matched or protected observation. Released MOSS therefore remains 7/21 bound and 14 blocking, the authoritative panel remains 55/126 bound and 71/126 blocking, and readiness remains zero of six. A rights-complete probe image and a passing constrained runtime receipt are positive evidence for those exact obligations only; they are not MOSS performance, H1–H4 evidence or comparative superiority.

The widest defensible contribution is therefore an identification-and-authority envelope, not a performance result for self-improvement: when one evidence fibre contains worlds requiring different minimal authorized revisions or promotion dispositions, optimization over that interface cannot recover the missing distinction. Only licensed acquisitions can refine the fibre, and authority-bearing evidence must remain outside candidate write custody. The finite controlled-state frontier makes acquisition cost, legal intervention state and fixed-versus-rectangular ambiguity explicit under known kernels; it does not certify any real kernel, revision taxonomy or development advantage.

The scientific question that remains is whether this composition improves fresh development outcomes while reducing harmful repair, evaluator gaming, false broad revision and false promotion, relative to matched alternatives. Until a campaign executes under complete protected identities and independent verification, transferable protected fresh-task improvement over matched self-edit and self-evolution baselines remains undetermined — which is a different statement from its having been determined to be false, and it is recorded and reported separately.

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